

Circuit quantization from first principles

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Superconducting circuit quantization conventionally starts from classical Euler-Lagrange circuit equations of motion. Invoking the correspondence principle yields a canonically quantized description of circuit dynamics over a bosonic Hilbert space. This approach has been very successful for describing experiments, but implicitly starts from the classical Ginsberg-Landau mean-field theory for the circuit. Here, we employ a different approach that starts from a microscopic fermionic Hamiltonian for interacting electrons, whose ground space is described by the Bardeen-Cooper-Schrieffer (BCS) many-body wave function that underpins conventional superconductivity. We introduce the BCS ground space as a subspace of the full fermionic Hilbert space, and show that projecting the electronic Hamiltonian onto this subspace yields the standard Hamiltonian terms for Josephson junctions, capacitors, and inductors, from which standard quantized circuit models follow. This approach does not impose a spontaneously broken symmetry so that it consistently describes quantized circuits that support superpositions of phases, and the canonical commutation relations between phase and charge are derived from the underlying fermionic commutation properties, rather than imposed. By expanding the projective subspace, this approach can be extended to describe phenomena outside the BCS ground space, including quasiparticle excitations.

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I. INTRODUCTION

Superconducting devices are one of the leading candidates for building quantum computers, showing great flexibility for implementing qubits and gates [1,2]. The low working temperature reduces the noise, providing for long coherence time [3]. The strong coupling between superconducting qubits and electromagnetic fields allows us to control and operate the system [4].

The conventional approach to quantizing superconducting circuits [5,6] starts with the classical circuit equations of motion given by Kirchhoff's laws, which are derived from conservation laws and constitutive relations for the circuit. For conservative circuits, these equations are the Euler-Lagrange equations, which can be generated from a Lagrangian. Canonical quantization then proceeds, via the correspondence principle, by postulating commutation relations between classically conjugate coordinates describing the circuit phase space.

These classical equations describe the dynamics of the Ginsburg-Landau (GL) superconducting order parameters ϕ_j defined at the nodes j of the circuit. The GL order parameter is itself a classical mean field, approximating an underlying

microscopic electronic Hamiltonian, and the (re)quantization procedure of the classical Euler-Lagrange equations for ϕ_j , described above, yields the conventional bosonic theory for $\hat{\phi}_j$ with the conjugate coordinates being the circuit charges $\hat{n}_j$.

This standard approach, which is illustrated in Fig. 1, is practically satisfactory for modeling the vast majority of experiments; however, it is conceptually unfulfilling for several reasons. First, we already have a very successful microscopic quantum theory of superconductivity, namely, Bardeen-Cooper-Schrieffer (BCS) theory, which describes the dynamics of the microscopic electronic Hilbert space of interacting electrons in a metal. As such, there should be a “direct” pathway from the microscopic electronic Hamiltonian to the circuit Hamiltonian, without leaving the fermionic Hilbert space. In this goal, there is a close analogy in modeling low-dimensional semiconducting systems (such as quantum dots [7] or quantum Hall fluids [8,9]), in which the microscopic Fermi-liquid theory of a semiconductor is projected onto a low-dimensional subspace spanned by the low-energy bound states of a spatially inhomogeneous, weakly confining potential landscape, yielding effective descriptions in terms of mass-renormalized, few-electron (or hole) systems [8].

Second, GL mean-field theory (MFT) does not typically include phenomena such as quasiparticles or quantum fluctuations in the gap parameter Δ , which are otherwise added “by hand” after requantization [10,11].

Third, when requantizing GL mean-field theory, the bosonic canonical commutation relation $[\hat{\phi}, \hat{n}] = i$ is postulated [12–14], or see Ref. [15] for variations in this approach. However, it is not clear that this is consistent with the Stone-von Neumann theorem when the spectrum of $\hat{\phi}$ has compact support on the interval $[-\pi, \pi)$, and the spectrum

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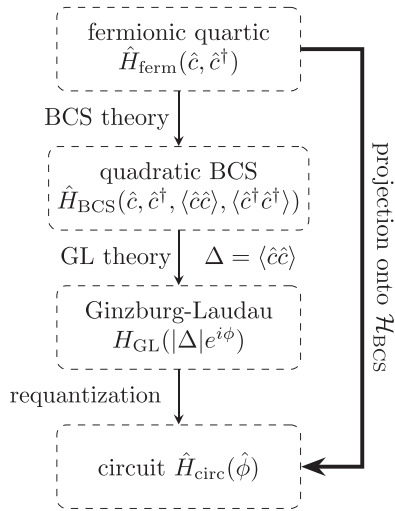


FIG. 1. The conventional approach to circuit quantization, along the left pathway, conceptually begins with a microscopic electronic theory of interacting fermions in a metal, which proceeds via a BCS ground-state ansatz to a classical Ginsburg-Landau mean-field theory (GL MFT) of superconductivity in a metal characterized by the complex-valued MFT order parameter $\Delta = |\Delta|e^{i\phi}$, which is finally requantized to produce a quantum theory of a circuit. The approach described here (right pathway) starts at the same point but uses the GL MFT as inspiration to define a low-energy BCS subspace $\mathcal{H}_{\text{BCS}}$ [see Eq. (13)], and then directly projects the dynamics onto this subspace, resulting in the effective low-energy circuit Hamiltonian.

of $\hat{n}$ is $\mathbb{Z}$. In fact, we expect that the commutation relations should be derived from the underlying fermionic degrees of freedom.

Fourth, GL mean-field theory assumes spontaneous symmetry breaking of the superconducting order parameter. However in the quantized circuit, we expect coherent quantum superpositions of (relative) phases [16], which is conceptually at odds with the spontaneous realization of only one among the large manifold of allowed phases.

Lastly, the classical GL MFT for the order parameter ϕ is indifferent to the support of ϕ : Descriptions in which ϕ is compact, $\phi \in [-\pi, \pi)$, and noncompact, $\phi \in \mathbb{R}$, are entirely equivalent at a classical level. In contrast, the quantum descriptions of these two alternatives, or anything in between, are not equivalent to one another [17]. Starting from the classical GL MFT therefore cannot resolve which alternative provides the more accurate description, whereas a microscopic derivation should.

The objective of this work is to provide a somewhat formal derivation of the Hamiltonian for superconducting circuits, starting from the microscopic theory of interacting electrons. To do this, we reformulate BCS theory slightly, by introducing a low-energy Hilbert space of the set of BCS ground states, $\mathcal{H}_{\text{BCS}} = \text{span}\{|\Psi(\phi_j)\rangle\}_{\phi_j \in \mathbb{Z}[-\pi, \pi)}$, where $|\Psi(\phi_j)\rangle$ is a BCS ground state (defined later), parameterized by a BCS phase angle ϕ_j . We define a projector P onto $\mathcal{H}_{\text{BCS}}$, and use this projector to directly construct the low-energy circuit Hamiltonian from the microscopic electronic Hamiltonian, as illustrated in Fig. 1.

We note that our projector-based approach to circuit quantization has some similarity to early work by Leggett [18] and Ambegaokar *et al.* [19], which evaluates expectations of microscopic interaction terms directly. The underlying analytical approach we take has a similar starting point to the early work by Widom *et al.* [12,13] and recently by Mizel [16], starting with a low-energy vector space defined by the BCS ground states.

We apply this projector-based approach to re-analyze the conventional superconducting toolbox: capacitors, inductors, and Josephson junctions, and we show that we recover the expected results in the limit of large systems.

In what follows, we formalize this approach, and derive a number of features of the model, including the emergent, discrete basis for the compact Hilbert subspace $\mathcal{H}_{\text{BCS}}$, the corresponding dimensionality of the low-energy subspace, and the canonical commutation relations, and we identify the relevant low-energy Hilbert space of standard lumped-element superconducting components. We specifically address the difference between the effective Hilbert space for inductors and Josephson junctions. With this approach, we describe the connection between the microscopic BCS theory of superconductors [20] and the mesoscopic electronic description of a Cooper-pair box or transmon [21], as well as devices that include an inductive element.

This paper is organized as follows. In Sec. II, we compute the overlap of BCS states and discuss the implications of this for the low-energy Hilbert space defined by these states. This section includes some of the main contributions of this work, in which we derive the *effective dimensionality* of the low-energy BCS subspace, we define a projector onto the BCS subspace, and we derive the commutator of the effective phase and charge operators. In Sec. III, we apply the low-energy BCS projector to derive the effective theory of a superconducting electronic LCJ circuit from the microscopic fermionic description of the system. This analysis provides a detailed derivation of geometric and kinetic contributions to the inductance, and provides an analysis of how the extended support of the phase parameter emerges from the microscopic description of an inductive wire. Finally we discuss some implications that follow from this analysis, and conclude with potential directions for further work.

A. Microscopic electronic Hamiltonian

To introduce notation, we consider two pointlike pieces of superconducting metal separated by an insulator, and connected by an extended wire, corresponding to the elements of an LCJ circuit shown in Fig. 2. For the Josephson junction illustrated in Fig. 2(b), the superconducting islands A and B are described by their respective Hamiltonians [22],

$$H_{A,B} = H_{K_{A,B}} + H_{I_{A,B}}, \quad (1)$$

where the microscopic electron kinetic and interaction terms are given, respectively, in the momentum basis as

$$H_{K_A} = \sum_{k_A \sigma_A} (\epsilon_{k_A} - \mu_A) c_{k_A \sigma_A}^\dagger c_{k_A \sigma_A}, \quad (2)$$

$$H_{I_A} = -|g|^2 \sum_{k_A k'_A} c_{k_A \uparrow}^\dagger c_{-k_A \downarrow}^\dagger c_{-k'_A \downarrow} c_{k'_A \uparrow}, \quad (3)$$

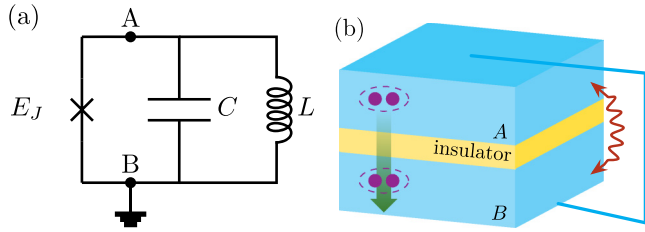


FIG. 2. (a) A standard LCJ circuit consists of an inductor L , a capacitor C , and a Josephson junction E_J made by superconductors. (b) A cartoon of the microscopic elements of a Josephson junction, consisting of two superconducting islands (blue) separated by an insulating tunnel barrier (yellow). At low temperature, the electrons bind into Cooper pairs that tunnel coherently across the insulating barrier. In addition, the charges inside the two superconducting islands have a capacitive Coulomb interaction (red arrow). We connect the two superconducting islands by a superconducting line (blue), in which the superconducting phase is spatially varying.

and similarly for H_B , where ϵ_k is the single-particle kinetic energy of an electron with momentum k , μ is the chemical potential, σ labels the spin index, g is the effective coupling strength between electrons, and $c_x^\dagger$ and c_x are creation and annihilation operators for electron mode x , respectively. We include a chemical potential in Eq. (2) to preempt the fact that in superconducting circuits, superconducting islands will (typically) be galvanically or tunnel coupled to one another, and eventually to ground, so that charge may be exchanged between parts of the circuit and to the electrical ground. For most transmon scenarios, the Josephson junctions are only connected to the electrical ground, which serves as a reservoir of charge [23]. The chemical potential here accounts for the soft constraint on charge conservation on an island in a grounded circuit.

Assuming pointlike metallic islands, the Coulomb interaction between electrons on islands A and B is given by

$$H_{\text{Coul}} = \lambda_C \sum_{k_A \sigma_A} c_{k_A \sigma_A}^\dagger c_{k_A \sigma_A} \sum_{k_B \sigma_B} c_{k_B \sigma_B}^\dagger c_{k_B \sigma_B} = 4\lambda_C \hat{N}_A^{\text{CP}} \hat{N}_B^{\text{CP}}, \quad (4)$$

where we define the “Cooper-pair number” operator

$$\hat{N}^{\text{CP}} = \hat{N}^{\text{el}}/2, \quad (5)$$

in terms of the electron-number counting operator $\hat{N}^{\text{el}} = \sum_{k\sigma} c_{k\sigma}^\dagger c_{k\sigma}$, and λ_C is the effective Coulomb interaction strength between the two pointlike superconductors.

The electrons also tunnel through the insulating barrier, with tunneling Hamiltonian

$$H_T = \sum_{k_A k_B \sigma_A \sigma_B} t_{k_A k_B} c_{k_A \sigma_A}^\dagger c_{k_B \sigma_B} + \text{H.c.}, \quad (6)$$

where $t_{k_A k_B}$ is the tunneling matrix element between modes in the different islands. The microscopic Hamiltonian for an electrical circuit including tunnel junctions is then

$$H_{\text{el}} = H_A + H_B + H_{\text{Coul}} + H_T. \quad (7)$$

For simplicity, we first approximate the islands A and B to be pointlike, such that the model is the standard CJ circuit. In

Sec. III D, we allow for an extended superconducting inductive wire galvanically connecting A and B , in which $H_A + H_B$ also includes kinetic current flow between the islands, giving rise to inductive terms in the Hamiltonian for an LCJ circuit, pictured in Fig. 2(b).

Conventionally, the quartic Hamiltonian in Eq. (1) describing each metallic island is treated approximately by factoring into a quadratic Hamiltonian with coefficients computed in mean-field theory, yielding

$$H_{\text{BCS}} = \sum_{ks} (\epsilon_k - \mu) c_{ks}^\dagger c_{ks} - |g|^2 \sum_{kk'} (\langle c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rangle c_{-k'\downarrow} c_{k'\uparrow} + \text{H.c.}), \quad (8)$$

$$= \sum_{ks} (\epsilon_k - \mu) c_{ks}^\dagger c_{ks} - \sum_k (\Delta^* c_{-k\downarrow} c_{k\uparrow} + \text{H.c.}), \quad (9)$$

where $\Delta = |g|^2 \sum_k \langle c_{-k\downarrow} c_{k\uparrow} \rangle = |\Delta| e^{i\phi}$ is the well-known complex-valued superconducting order parameter. The quadratic Hamiltonian is diagonalized using the Bogoliubov-Valatin transformation to construct the BCS ground state:

$$|\Psi(\Delta)\rangle = \prod_k (u_k(|\Delta|) + v_k(|\Delta|) e^{i\phi} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle, \quad (10)$$

where $u_k, v_k \in [0, 1]$ are variational parameters used to minimize the total energy, $|0\rangle$ is the vacuum of the electronic Fock space, and $\phi \equiv \arg(\Delta) \in [-\pi, \pi]$ is an undetermined phase angle [22,24]. The self-consistent mean-field theory is found by requiring that the expectation in the definition of Δ is taken with respect to Eq. (10). The Bogoliubov coefficients that minimize the self-consistent energy of Eq. (8) are given by

$$u_k^2 = (1 + (\epsilon_k - \mu)/E_k)/2, \quad v_k^2 = (1 - (\epsilon_k - \mu)/E_k)/2, \quad (11)$$

where $E_k \equiv \sqrt{(\epsilon_k - \mu)^2 + \Delta^2}$ [22].

We use this standard approach to motivate a choice of low-energy basis states, but depart from it in that we do not impose the symmetry breaking implied by taking the partial expectations that transforms the microscopic quartic Hamiltonian in Eq. (1) into the quadratic form in Eq. (8).

We comment on our choice to use BCS’s parameterized wave functions to define an effective, low-energy subspace of the full fermionic Hilbert space on which Eq. (7) acts. Other choices may be possible, for example, the exact solution to the pairing Hamiltonian given by Richardson [25]. However, while Ref. [25] gives a formally exact wave function for pairing Hamiltonians of isolated fermionic systems in which the kinetic and interaction terms commute, practical application of this approach requires the solution of a set of coupled nonlinear algebraic equations, the multinomical order of which scales with the number of Cooper pairs. This makes it practically unwieldy for analyzing mesoscopic superconducting circuits, particularly those in which the electron number fluctuates through tunneling processes.

In any case, the BCS wave function and Richardson’s solution [25] are both wave functions defined on the underlying fermionic Hilbert space, and are each plausible

choices with which to define the low-energy subspace relevant to tunnel-coupled superconducting islands making up a superconducting circuit. A similar projector-based approach to deriving circuit theory starting from Richardson's solution would mirror the approach described here, replacing the BCS basis with a corresponding Richardson basis. Given the quantitative success of BCS theory in the analysis of superconductivity, for example, the application of the Ambegaokar-Baratoff relation in experimental calibration [26], and the agreement between the theories of BCS and Richardson in suitable limits [27,28], we expect any differences arising from either choice would be minor.

II. LOW-ENERGY BCS SUBSPACE

In this section, we use the phase-symmetry-broken BCS ground state $|\Psi(\Delta)\rangle$ to motivate the introduction of a low-energy Hilbert space spanned by a set of linearly independent BCS ground states $\{|\Psi(\phi)\rangle\}_{\phi \in [-\pi, \pi]}$.

To understand the structure of this Hilbert space, we analyze the overlap between BCS states from this set and determine conditions under which we can construct an approximately orthogonal and complete basis. We then construct ground-space phase and number operators, and then compute the phase-number commutator, which we connect to the well-known Pegg-Barnett theory [29].

A. BCS ground space

Motivated by Eq. (10), we define a family of BCS-type ground-state wave functions for a superconducting island as [16,20]

$$|\Psi(\phi_j)\rangle = \prod_k (u_k + v_k e^{i\phi_j} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle, \quad (12)$$

where $\phi_j \in [-\pi, \pi)$ parameterizes the allowed set of BCS states. We adopt the subscript j to preempt our intent to restrict the allowed phases to a discrete subset $\mathcal{Z} = \{\phi_j\}_{j \in \mathbb{Z}_d}$, where d is the effective dimension of the Hilbert space, which we will compute below.

The core of our approach here is to define a basis of BCS ground states $\{|\Psi(\phi_j)\rangle\}_{\phi_j \in \mathcal{Z}}$ and the corresponding low-energy Hilbert space:

$$\mathcal{H}_{\text{BCS}} = \text{span}\{|\Psi(\phi_j)\rangle\}_{\phi_j \in \mathcal{Z}}. \quad (13)$$

Projecting the microscopic electronic Hamiltonian into this subspace allows us to construct the low-energy circuit theory without passing through the conventional GL MFT followed by a requantization procedure.

For a system consisting of multiple superconducting islands, the total Hilbert space is formed from a tensor product of the island subsystems. For example, the state of a system consisting of metallic islands A and B is spanned by the tensor product of single-island states as

$$\begin{aligned} |\Psi_{AB}(\phi_A, \phi_B)\rangle &= |\Psi_A(\phi_A)\rangle \otimes |\Psi_B(\phi_B)\rangle \\ &= \prod_{k_A, k_B} (u_{k_A} + v_{k_A} e^{i\phi_A} c_{k_A\uparrow}^\dagger c_{-k_A\downarrow}^\dagger) \\ &\quad \times (u_{k_B} + v_{k_B} e^{i\phi_B} c_{k_B\uparrow}^\dagger c_{-k_B\downarrow}^\dagger) |00\rangle \end{aligned}$$

$$\begin{aligned} &= \prod_{k_A, k_B} (u_{k_A} + v_{k_A} e^{i\phi_{AB}} c_{k_A\uparrow}^\dagger c_{-k_A\downarrow}^\dagger) \\ &\quad \times (u_{k_B} + v_{k_B} e^{i\phi_{AB}} c_{k_B\uparrow}^\dagger c_{-k_B\downarrow}^\dagger) |00\rangle \\ &= |\Psi_{AB}(\phi_{AB})\rangle, \end{aligned} \quad (14)$$

where $|00\rangle$ denotes the electron vacuum of the two islands, and a unitary transformation on the electronic mode operators allows us to express the joint state as a function of the relative phase between the two superconducting islands $\phi_{AB} \equiv \phi_A - \phi_B$.

B. BCS ground-space projection operator

This work is focused on the dynamics of an electronic system restricted to the BCS ground space, so we define an (approximate) BCS ground-space projection operator P onto $\mathcal{H}_{\text{BCS}}$ [30] as

$$P = \sum_j |\Psi(\phi_j)\rangle \langle \Psi(\phi_j)|, \quad (15)$$

with the complementary projector defined as $\bar{P} = \mathbb{I} - P$. The degree to which this is a well-defined, idempotent projector satisfying $P^2 = P$ depends on the orthogonality of the low-energy phase basis, which we address below.

We will use the projection operator to construct effective operators acting on the low-energy space. For example, in a system of two superconducting islands with a microscopic electronic Hamiltonian consisting of intra-island and inter-island Coulomb and tunneling terms given by Eq. (7), the low-energy effective Hamiltonian is given by

$$PH_{\text{eff}}P \approx PH_0P + PH_1P + PH_1\bar{P}(\bar{P}E_0 - \bar{P}H_0\bar{P})^{-1}\bar{P}H_1P, \quad (16)$$

up to the second order in the tunneling, where $H_0 = H_A + H_B + H_{\text{Coul}}$ and $H_1 = H_T$ is treated to second order in perturbation theory [19,30].

C. BCS subspace orthogonality and completeness

The overlap of two BCS states $|\Psi(\phi)\rangle$ and $|\Psi(\phi')\rangle$ is given by

$$\mathcal{W}(\varphi) \equiv \langle \Psi(\phi) | \Psi(\phi') \rangle = \prod_k (u_k^2 + v_k^2 e^{i\varphi}), \quad (17)$$

where $\varphi \equiv \phi' - \phi$. Such overlaps have been considered by Onishi and Yoshida [31], and are discussed in Ref. [16].

To characterize the functional behavior of $\mathcal{W}$, we assume a symmetric electronic band consisting of $n \gg 1$ single-particle modes with bandwidth $\pm\mathcal{B}$ at half filling, $N_{\text{el}} = n/2$, so that $-\mathcal{B} \leq \epsilon_k \leq \mathcal{B}$ and $\mu = 0$. We nondimensionalize the bandwidth relative to the superconducting gap as $b = \mathcal{B}/\Delta$.

From BCS theory [22], $\Delta = 2\mathcal{B}e^{-1/\lambda}$, where the dimensionless quantity $\lambda = |g|^2 n / (2\mathcal{B}) \ll 1$ is the electron-phonon coupling strength in units of the single-particle density states (which we rederive in Sec. III A). It follows that

$$b = \mathcal{B}/\Delta = e^{1/\lambda} / 2 \gg 1. \quad (18)$$

In typical metallic systems, $\mathcal{B} \gtrsim 1$ eV [32] and $\Delta \sim 1$ meV [33], so $b > 1000 \gg 1$.

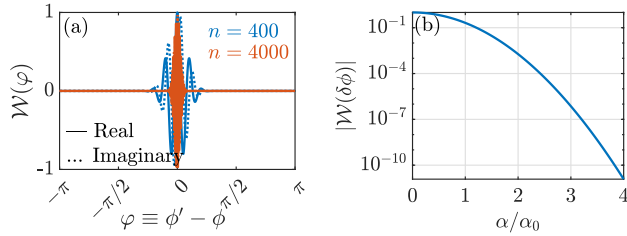


FIG. 3. (a) Overlap of the BCS states $\mathcal{W}(\varphi) = \langle \Psi(\phi) | \Psi(\phi') \rangle$ as a function of the phase difference $\varphi \equiv \phi' - \phi$. Increasing the number of electron modes, n , leads to faster oscillations over a narrower range, and the overlap $\mathcal{W}(\varphi)$ tends to a pseudo-Kronecker delta function $\mathcal{W}(\phi_j - \phi_{j'}) \xrightarrow{n \rightarrow \infty} \delta_{jj'}$. For this plot, we have set $b = 10$. (b) The overlap amplitude of adjacent basis states given in Eq. (24). As α grows, neighboring phase states become increasingly orthogonal.

We take the logarithm of Eq. (17) and replace the resulting momentum sum with an integral over the Bogoliubov energies $E = \pm\sqrt{\epsilon^2 + \Delta^2}$, i.e., $\sum_k \bullet \rightarrow \int_{-B}^B dE \rho(E) \bullet$, where the density of Bogoliubov energies is given by [34]

$$\rho(E) = \frac{n}{2\Delta\sqrt{b^2 - 1}} \frac{|E|}{\sqrt{E^2 - \Delta^2}} \Theta(|E| - \Delta), \quad (19)$$

and Θ is the unit-step function. We then find

$$\begin{aligned} \ln \mathcal{W} &= \sum_k \ln \left(\frac{1}{2} \left(1 + \frac{\epsilon_k}{E_k} \right) + \frac{1}{2} e^{i\varphi} \left(1 - \frac{\epsilon_k}{E_k} \right) \right) \\ &= \int_{-B}^B dE \rho(E) \ln \left(\frac{1}{2} \left(1 + \frac{\epsilon_E}{E} \right) + \frac{1}{2} e^{i\varphi} \left(1 - \frac{\epsilon_E}{E} \right) \right) \\ &= \int_{\Delta}^{b\Delta} dE \rho(E) \ln \left(e^{i\varphi} \left(1 - \frac{\Delta^2}{2E^2} (1 - \cos \varphi) \right) \right) \\ &= n(i\varphi/2 - \pi \sin(\varphi/4)^2/b) + O(b^{-2}) \\ &= in\varphi/2 - \pi n\varphi^2/(16b) + O(\varphi^3, b^{-2}), \end{aligned} \quad (20)$$

where $\epsilon_E = \sqrt{E^2 - \Delta^2}$, and we have retained the leading terms in b^{-1} . It follows that

$$\mathcal{W}(\varphi) \approx e^{in\varphi/2 - \pi n\varphi^2/(16b)}, \quad (21)$$

so that the overlap is approximately a Gaussian envelope of width $\sim \sqrt{b/n}$, modulated by an oscillating function. We plot $\mathcal{W}(\varphi)$ in Fig. 3(a), showing that $\mathcal{W}(0) = 1$, and that $|\mathcal{W}|$ decreases quickly for $\varphi > \sqrt{b/n}$.

We now use the functional form of $\mathcal{W}$ to establish conditions under which we can construct an (approximately) orthogonal basis of phase states that is also (approximately) complete in the low-energy subspace.

1. Basis orthogonality

An orthogonal basis of phase states would satisfy $\mathcal{W}(\phi_j - \phi_{j'}) = \delta_{jj'}$ [18]. However, $|\mathcal{W}(\varphi)| > 0$ everywhere, and so any set $\{|\Psi(\phi)\rangle\}$ is not orthogonalized. Nonetheless, Fig. 3(a) shows that $|\mathcal{W}(\varphi)|$ is a rapidly decreasing function of $|\varphi|$, and so we construct an approximately orthogonal basis by discretizing $\phi \in [-\pi, \pi)$ with discretization interval $\delta\phi$, to construct the discrete set of phases $\phi_j = -\pi + \delta\phi j$ with

$j \in \mathbb{Z}_d \equiv \{0, 1, \dots, d = \lfloor 2\pi/\delta\phi \rfloor\}$, so that

$$\phi_j \in \mathcal{Z} \equiv 2\pi(\mathbb{Z}_d/d - 1/2) \subset [-\pi, \pi). \quad (22)$$

Equation (21) provides a foundation on which to choose the discretization interval. The overlap amplitude of successive basis states is given by

$$|\mathcal{W}(\phi_{j+1} - \phi_j)| = |\mathcal{W}(\delta\phi)| = e^{-\delta\phi^2 n \pi / (16b)}. \quad (23)$$

We define $\alpha = \delta\phi\sqrt{n}/(2\pi)$ and $\alpha_0 = \sqrt{2b}/\pi$ so that

$$|\mathcal{W}(\delta\phi)| = e^{-\pi^3 \alpha^2 / (4b)} = e^{-\pi \alpha^2 / 2\alpha_0^2} \quad (24)$$

is nominally independent of n . With these definitions, the effective dimension of the BCS subspace is $d = 2\pi/\delta\phi = \sqrt{n}/\alpha$, and the set $\mathcal{Z}$ becomes dense in the interval $[-\pi, \pi)$ as n grows. In addition, $d \ll n$, so that the effective dimension of the BCS subspace is much smaller than the number of single-particle modes in the metallic island.

The ratio α/α_0 controls the approximate orthogonality of the BCS phase basis discretization: Basis orthogonality improves with larger α . From Fig. 3(b), we see that choosing $\alpha \gtrsim 2\alpha_0$ produces a nearly orthogonal basis with $|\mathcal{W}(\delta\phi)| < e^{-2\pi} \sim 10^{-3}$. With this choice, we have the discretization interval and effective Hilbert space dimension given, respectively, by

$$\delta\phi = 4\sqrt{2b/n}, \quad (25)$$

$$d_{\text{eff}} = \frac{\pi}{2} \sqrt{\frac{n}{2b}}. \quad (26)$$

This formalizes the assertion [18] that distinct BCS phase states can be approximated as being orthogonal in a suitable limit, and provides an estimate for the effective, coarse-grained dimensionality of the phase basis, which was also addressed by Mizel [16].

In Sec. IV A 1, we will see that for some typical junction parameters, $n/(2b) \sim 10^3$, so that $d_{\text{eff}} \sim 50$ is the (surprisingly small) effective Hilbert space dimension of a pointlike superconducting element. This scale is similar to the number of microscopic constituents that have been estimated to participate in superposition states in small, experimentally accessible flux qubits [35].

2. Basis completeness

Given a discrete set of phases $\mathcal{Z}$, we cannot perfectly represent an arbitrary state $|\Psi(\phi)\rangle$ where $\phi \notin \mathcal{Z}$. Here, we analyze how well the Hilbert space $\mathcal{H}_{\text{BCS}} = \text{span}\{|\Psi(\phi_j)\rangle\}_{\phi_j \in \mathcal{Z}}$ covers all possible phase states $|\Psi(\phi)\rangle$ by bounding the distance d between $|\Psi(\phi)\rangle$ and $\mathcal{H}_{\text{BCS}}$. We find

$$\begin{aligned} d(|\Psi(\phi)\rangle, \mathcal{H}_{\text{BCS}}) &= \min_{\{a_j\}} d \left(|\Psi(\phi)\rangle, \sum_j a_j |\Psi(\phi_j)\rangle \right) \\ &= \min_{\{a_j\}} \left| |\Psi(\phi)\rangle - \sum_j a_j |\Psi(\phi_j)\rangle \right| \\ &\leq \min_{\{a_j\}} \sqrt{2(1 - \sum_j a_j |\mathcal{W}(\phi_j - \phi)|)} \end{aligned}$$

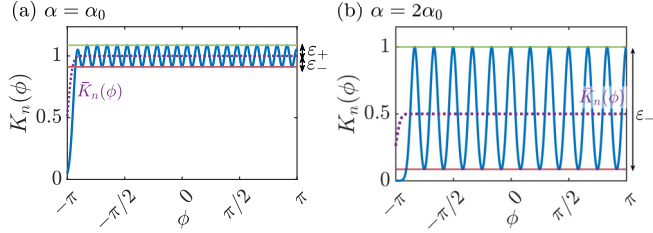


FIG. 4. Projections K_n (oscillatory lines) and their integral approximations $\bar{K}_n$ (dotted lines) for (a) $\alpha = \alpha_0$, in which the approximate BCS projection is close to unity everywhere but is a strongly nonorthogonal basis with $K_n > 1$, and (b) $\alpha = 2\alpha_0$, where the basis is very nearly orthogonal but with poorly represented states at $\phi = (\phi_{j+1} - \phi_j)/2$. The parameters are chosen to be $n = 10^3$ and $b = 10$.

$$\begin{aligned} &\leq \sqrt{2 \min_j (1 - |\mathcal{W}(\phi_j - \phi)|)} \\ &\leq \sqrt{2(1 - |\mathcal{W}(\delta\phi)|)} \\ &\approx \sqrt{2(1 - e^{-\pi\alpha^2/(2\alpha_0^2)})}. \end{aligned} \quad (27)$$

For $\alpha \ll \alpha_0$, we see that $d(|\Psi(\phi)\rangle, \mathcal{H}_{\text{BCS}}) \approx \sqrt{\pi}\alpha/\alpha_0$. That is, smaller values of α generate a basis that is capable of representing a more complete set of phases. This is contrary to the requirement of basis orthogonality, which favors larger α . There is therefore a trade-off between the orthogonality of the phase basis and the completeness of the representation.

3. Projection error

Given the trade-off between basis orthogonality and completeness when choosing the discretization parameter α , we synthesize both considerations by computing the error in projecting an arbitrary phase state onto the low-energy BCS subspace. We quantify the degree to which the discretized subspace $\mathcal{H}_{\text{BCS}}$ can represent an arbitrary phase state $|\Psi(\phi)\rangle$, by computing its projection onto $\mathcal{H}_{\text{BCS}}$:

$$K_n(\phi) \equiv \langle \Psi(\phi) | P | \Psi(\phi) \rangle = \sum_{\phi_j \in \mathcal{Z}} |\mathcal{W}(\phi - \phi_j)|^2. \quad (28)$$

This quantity is oscillatory, with maxima when $\phi \in \mathcal{Z}$, and minima when $\phi = (\phi_j + \phi_{j+1})/2$, and is plotted in Fig. 4 for $\alpha = \alpha_0$ and for $\alpha = 2\alpha_0$. The mean of the oscillatory projection is approximated by turning the sum into an integral,

$$\begin{aligned} \bar{K}_n(\phi) &= \int_0^d d\phi j e^{-n(\phi + \pi - \delta\phi_j)^2/(4\alpha_0^2)} \\ &\approx \frac{\alpha_0}{2\alpha} \left(\text{erf} \left[\frac{\sqrt{n}(\pi + \phi)}{2\alpha_0\sqrt{\pi}} \right] \right. \\ &\quad \left. - \text{erf} \left[\frac{\sqrt{n}(\pi + \phi) - 2\pi\sqrt{n-1}}{2\alpha_0\sqrt{\pi}} \right] \right), \end{aligned} \quad (29)$$

which averages the maxima and minima of K_n . The integral approximation $\bar{K}_n$ is also shown in Fig. 4.

For small α , $K_n \sim 1$ everywhere indicating that it can well represent arbitrary phase states; however, $K_n > 1$ for some values of ϕ indicating that it is not well orthogonalized. Con-

versely, for larger α , the minima in K_n are much lower than unity, indicating that the subspace cannot well represent arbitrary phase states. The maxima and minima of K reflect the trade-off between completeness and orthogonality, described above. We conclude that $\alpha \approx 2\alpha_0$ is a reasonable trade-off between basis completeness and orthogonality.

For very small metallic islands where $n \sim b$, there may be observable physical consequences arising from small n effects [28], but we do not pursue this analysis any further here.

4. Orthonormalization and effective dimensionality

The choice $\alpha = 2\alpha_0$ was somewhat heuristic, so we provide some additional analysis that gives an independent estimate for the effective dimensionality of the phase basis $\mathcal{d}_{\text{eff}}$. The approach starts with a very large phase-basis set $\mathcal{Z}$, diagonalizes the phase-basis overlap matrix, and then computes the effective, reduced dimensionality of the Hilbert space.

As discussed above, the phase basis is normalized but not orthonormalized. Given an initial choice of the phase-basis size $|\mathcal{Z}| = d$, we define the $d \times d$ overlap matrix $\underline{W}$ with matrix elements $W_{lm} = \langle \Psi(\phi_l) | \Psi(\phi_m) \rangle = \mathcal{W}(\phi_l - \phi_m)$ so that

$$\underline{W} = \begin{bmatrix} \mathcal{W}(0) & \mathcal{W}(-\delta\phi) & \dots \\ \mathcal{W}(\delta\phi) & \mathcal{W}(0) & \dots \\ \mathcal{W}(2\delta\phi) & \mathcal{W}(\delta\phi) & \dots \\ \vdots & \vdots & \vdots \end{bmatrix}. \quad (30)$$

Since $\mathcal{W}(-x) = \mathcal{W}(x)^*$, $\underline{W}$ is Hermitian, so it is unitarily diagonalizable with real eigenvalues. Further, it is a circulant matrix, in which each column vector is a cyclic permutation of its predecessor, and so $\underline{W}$ is generated by its first column vector $\underline{w}$, i.e., $\underline{W} = [\underline{w}, \pi_1 \underline{w}, \pi_2 \underline{w}, \dots]$, where π_j is the cyclic permutation by j slots. Diagonalizing $\underline{W}$ provides a linear transformation that orthogonalizes the original phase basis.

It is important to note that $\underline{W}$ is invertible for all d , so its matrix rank is d ; that is, formally the Hilbert space dimension of the phase basis is always given by d . However, for large $d \gtrsim \sqrt{n/b}$, a phase-basis state becomes highly overlapped with its neighboring states, reflected in the fact that $\underline{W}$ becomes increasingly ill-conditioned. So, there is a sense in which the effective dimensionality of the Hilbert space defined by $\mathcal{Z}$ can be much smaller than d . We address this here.

Circulant matrices are diagonalized by the unitary discrete Fourier transform matrix $\underline{F}$ [16] whose matrix elements are $F_{lm} = e^{2\pi i(l-1)(m-1)/d}/\sqrt{d}$, so

$$\underline{W} = \underline{F}^\dagger \cdot \underline{D} \cdot \underline{F}, \quad (31)$$

where $\underline{D} = \text{diag}(\underline{\Lambda})$ is a diagonal matrix of real and positive eigenvalues. In addition, for a $d \times d$ circulant matrix, the eigenvalues are proportional to the discrete Fourier transform of the first column vector,

$$\underline{\Lambda} = \sqrt{d} \underline{F} \cdot \underline{w}, \quad (32)$$

which we compute below.

Representing a column vector of basis states as $|\phi\rangle = [\dots, |\Psi(\phi_j)\rangle, \dots]^T$, we seek a (nonunitary) basis transformation $\underline{E}$ that orthonormalizes the phase basis $\{|\Psi(\phi_j)\rangle\}$ into an

orthonormal basis $\{|e_j\rangle\}$, i.e.,

$$|e\rangle = \underline{E} \cdot |\phi\rangle, \quad (33)$$

with $\langle e_i | e_j \rangle = \sum_{lm} E_{il}^* W_{lm} E_{jm} = \delta_{ij}$. That is, $\underline{E}^* \cdot \underline{W} \cdot \underline{E}^\top = \mathbb{I}$. Noting that $\underline{F}^\dagger \cdot \underline{F} = \mathbb{I}$, $\underline{F} = \underline{F}^\top$, and $\underline{F}^* = \underline{F}^\dagger$, it is straightforward to verify that

$$\underline{E} = \underline{F} \cdot \underline{D}^{-1/2} \cdot \underline{F}^\dagger = (\underline{W}^\top)^{-1/2}. \quad (34)$$

Defining the Fourier-transformed bases $|\underline{f}\rangle = \underline{F}^\dagger \cdot |e\rangle$, which is orthonormal, and $|\underline{n}\rangle = \underline{F}^\dagger \cdot |\phi\rangle$, which is merely orthogonal, Eq. (33) becomes

$$|\underline{n}\rangle = \underline{D}^{1/2} \cdot |\underline{f}\rangle. \quad (35)$$

That is, $|n_k\rangle = \sqrt{\Lambda_k} |f_k\rangle$, so that the “number” basis satisfies $\langle n_k | n_q \rangle = \Lambda_k \delta_{kq}$. The Fourier transform matrix thus maps the nonorthogonal-but-normalized phase basis $|\phi\rangle$ into the orthogonal-but-unnormalized number basis $|\underline{n}\rangle$, where $\sqrt{\Lambda_k}$ determines the unnormalized length of the corresponding number state. The distinction between the orthogonal basis $|\underline{n}\rangle$ (which is unnormalized) and the orthonormal basis $|\underline{f}\rangle$ is significant, but not covered in some treatments (see, e.g., Chapter 3 of Ref. [36]).

The normalization $\sqrt{\Lambda_k}$ provides a measure of the relative “significance” of each number state derived from the Fourier transform of the original nonorthogonal phase basis: Those with large normalization have large support on the orthonormal basis $|\underline{f}\rangle$, while those with small normalization are relatively insignificant and can be truncated without much error. We therefore define the “relative significance” of $|n_k\rangle$ as $s_k = \sqrt{\Lambda_k / \Lambda_{\max}}$, where $\Lambda_{\max} = \max_k \Lambda_k$.

The relative significance allows us to compute the effective dimensionality of the Hilbert space defined by the phase basis. By analogy, suppose that $\underline{W}$ was idempotent with $s_k = 0$ or 1, defining an ideal projector. The dimension of the projected space would be

$$d_{\text{eff}} = \sum_k s_k. \quad (36)$$

We generalize this to be the definition of the “effective dimensionality” for any basis overlap matrix $\underline{W}$. Appendix C illustrates this for a two-dimensional (2D) toy example, exemplifying that this definition coincides with the usual integer-valued dimensionality for colinear or orthonormal bases. We now compute estimates for the dependence of d_{eff} on the choice of the basis size d .

In the limit $d \ll \sqrt{n/b}$, the phase discretization $\delta\phi = 2\pi/d$ will be “large” and $|\mathcal{W}(\delta\phi)| \ll 1$, so that the phase basis will itself be very close to an orthonormal basis. This is the scenario where the phase basis is relatively sparse on the interval $[-\pi, \pi)$, illustrated in Fig. 4(b), and poorly represents phase values where $K_n(\phi)$ is small. In this limit, $\underline{W} \approx \mathbb{I}$, so $\Lambda_k \approx 1$ for all k , and the effective dimensionality of the phase basis is $d_{\text{eff}} = \sum_k s_k \approx d$, as expected for a nearly orthonormal basis of size d . Since $d = |\mathcal{Z}|$ is essentially an arbitrary choice made for calculational purposes, in this small- d limit, d is not itself a physically significant property of the system.

Conversely, in the limit $d \gg \sqrt{n/b}$, the phase discretization $\delta\phi$ is relatively small, so the discrete phase basis can

well represent the continuum of phase values, but neighboring phase-basis states will strongly overlap one another, and $\underline{W}$ will be far from the identity. This is the scenario illustrated in Fig. 4(a). In this large- d limit, we use Eq. (32) and take the Fourier transform of Eq. (21) to estimate the eigenvalues:

$$\begin{aligned} \Lambda_k &= \sum_{m=0}^{d-1} e^{2\pi i k m} \mathcal{W}(m \delta\phi) \\ &\approx \frac{d}{2\pi} \int_{-\infty}^{\infty} d\phi e^{i\phi k} \mathcal{W}(\phi) \\ &= \frac{2d}{\pi} \sqrt{\frac{b}{n}} e^{-4b(k+n/2)^2/(\pi n)}, \end{aligned} \quad (37)$$

where $k \in \{-n/2 - d/2, \dots, -n/2 + d/2 - 1\} \cong \mathbb{Z}_d$, and in the second line we approximate $\mathcal{W}$ as negligibly small for $|\phi| > \pi$, which is a good approximation resulting from Eq. (21).

Thus, if d is chosen to be sufficiently large, then Λ_k is approximately Gaussian, with $\Lambda_{\max} = 2d\sqrt{b/n}/\pi$ at $k = -n/2$. The effective dimensionality is then

$$d_{\text{eff}} = \sum_k s_k \approx \int_{-\infty}^{\infty} dk s_k = \pi \sqrt{\frac{n}{2b}}. \quad (38)$$

This matches Eq. (26), up to a factor of 2.

Importantly, Eq. (38) is independent of the initial choice of the size of the phase-basis set d . In addition, through Eq. (18), d_{eff} depends only on physical parameters from the underlying microscopic electronic theory. That is, in the large- d limit, d_{eff} is a physically relevant quantity that emerges naturally from the structure of the low-energy BCS subspace.

Finally, we conclude this section by noting that Eq. (38) arises from the kinematics of the Hilbert space defined by the phase basis. It is not determined by energetic considerations, which require the introduction of low-energy dynamics that we discuss later.

D. Phase and charge operators

having constructed a low-energy Hilbert space from a discrete basis of phase states, in this section we define a basis of “number” states through the Fourier transform. These two bases implicitly define the low-energy phase and charge operators $\hat{\phi}$ and $\hat{N}$, and from these we compute commutation relations. For simplicity, we will assume that the phase basis is an orthonormal basis, notwithstanding the preceding discussion.

The phase operator is straightforwardly defined as

$$\hat{\phi} = \sum_{\phi_j \in \mathcal{Z}} \phi_j |\Psi(\phi_j)\rangle \langle \Psi(\phi_j)|, \quad (39)$$

recalling that $\phi_j = \phi_0 + \delta\phi j$, and we choose the convention $\phi_0 = -\pi$.

The BCS states $|\Psi(\phi_j)\rangle$ are not eigenstates of the electron number. Here, we define low-energy “number” eigenstates $|\tilde{\Psi}(N)\rangle$ via a discrete Fourier transform:

$$|\tilde{\Psi}(N)\rangle = \frac{1}{\sqrt{d}} \sum_{\phi_j \in \mathcal{Z}} e^{-iN\phi_j} |\Psi(\phi_j)\rangle. \quad (40)$$

We show in Appendix A that $|\tilde{\Psi}(N)\rangle$ are eigenstates of the Cooper-pair number operator defined in Eq. (5) so that

$$\hat{N}^{\text{CP}}|\tilde{\Psi}(N)\rangle = N|\tilde{\Psi}(N)\rangle, \quad (41)$$

establishing that the Fourier conjugate states are indeed charge-number eigenstates, and the set $\{|\tilde{\Psi}(N)\rangle\}$ defines a conjugate charge-number basis with $N \in \{n/2 - d/2, \dots, n/2 + d/2\} \subset \mathbb{Z}$. The center of this interval corresponds to half filling of the band. The number basis implicitly defines the number operator in the low-energy

subspace:

$$\hat{N} = P\hat{N}^{\text{CP}}P = \sum_{N=n/2-d/2}^{n/2+d/2} N|\tilde{\Psi}(N)\rangle\langle\tilde{\Psi}(N)|. \quad (42)$$

Pegg and Barnett have shown that the phase-number commutator follows the commutator-Poisson-bracket correspondence for physical states [29]. Here, we compute the canonical commutation relation between the matter phase operator $\hat{\phi}$ and number operator $\hat{N}$ in a single superconducting island; we then generalize to the relative phase of a system consisting of two superconductors.

We represent the phase operator $\hat{\phi}$ in the number basis:

$$\begin{aligned} \hat{\phi} &\equiv \sum_{j=0}^d \phi_j |\Psi(\phi_j)\rangle\langle\Psi(\phi_j)| = \phi_0 \mathbb{I} + \frac{2\pi\alpha}{\sqrt{n}} \sum_{j=0}^d \frac{j}{\sqrt{n-1} + \alpha} \sum_{NN'} e^{i(N-N')(\phi_0 + \frac{2\pi\alpha}{\sqrt{n}}j)} |\tilde{\Psi}(N)\rangle\langle\tilde{\Psi}(N')| \\ &= \left(\phi_0 + \frac{\pi\alpha}{\sqrt{n}}\right) \mathbb{I} + \frac{2\pi\alpha}{\sqrt{n}} \frac{\sqrt{n-1}}{\sqrt{n-1} + \alpha} \sum_{N \neq N'} e^{i\phi_0(N-N')} \frac{e^{2\pi i\alpha(N-N')/\sqrt{n-1}}}{e^{2\pi i\alpha(N-N')/\sqrt{n-1}-1}} |\tilde{\Psi}(N)\rangle\langle\tilde{\Psi}(N')|. \end{aligned} \quad (43)$$

In the number basis, we compute the phase-number commutator matrix elements, and for $n \gg |N' - N|$, we find

$$\langle\tilde{\Psi}(N)|[\hat{\phi}, \hat{N}]|\tilde{\Psi}(N')\rangle = \begin{cases} 0, & N = N', \\ \frac{2\pi\alpha\sqrt{n-1}}{\sqrt{n}(\sqrt{n-1} + \alpha)}(N' - N)e^{i\phi_0(N-N')} \frac{e^{2\pi i\alpha(N-N')/\sqrt{n-1}}}{e^{2\pi i\alpha(N-N')/\sqrt{n-1}-1}}, & N \neq N'. \end{cases} \quad (44)$$

For finite-dimensional operators, the commutator is traceless, and in the limit $n \rightarrow \infty$, we find

$$\langle\tilde{\Psi}(N)|[\hat{\phi}, \hat{N}]|\tilde{\Psi}(N')\rangle \approx i(1 - \delta_{NN'})e^{i\phi_0(N-N')}. \quad (45)$$

The ϕ_0 dependence is consistent with the results of Pegg-Barnett theory [29] for phase-number commutators. Transforming back to the ϕ basis, the commutator becomes

$$[\hat{\phi}, \hat{N}] \approx -i(\mathbb{I}_{d \times d} - d|\Psi(\phi_0)\rangle\langle\Psi(\phi_0)|), \quad (46)$$

where we have used the fact that $(\sqrt{n-1} + \alpha)/\alpha = 2\pi/\delta\phi = d$. Since the Hilbert space is finite dimensional, the commutator is traceless, as required by the Stone-von Neumann theorem for finite-dimensional operators. This commutator structure is notably different from $[\hat{\phi}, \hat{N}] = -i$ postulated in other circuit quantization procedures [12,14], or from commutation relations that are implicit in ϕ , e.g., as described by Bishop [37], $[e^{i\hat{\phi}}, \hat{N}] = e^{i\hat{\phi}}$, where $e^{i\hat{\phi}}$ represents the non-Hermitian ladder operator in the number basis. In contrast, in the commutator derived in Eq. (46), both $\hat{\phi}$ and $\hat{N}$ are Hermitian observables that appear on equal footing.

The expectation value of the commutator with respect to an arbitrary phase state $|\Psi(\phi)\rangle$ is given by

$$\begin{aligned} \langle\Psi(\phi)|[\hat{\phi}, \hat{N}]|\Psi(\phi)\rangle &\approx -i(1 - d|\mathcal{W}(\phi_0 - \phi)|^2) \\ &\approx -i(1 - d\delta_{\phi_0, \phi}). \end{aligned} \quad (47)$$

In Josephson junctions, consisting of two superconducting islands A and B , we are interested in the commutator of the phase and number differences between the two islands. Assuming the junction is symmetric such that $n_A \approx n_B \equiv n$ and

$\alpha_A \approx \alpha_B \equiv \alpha$, we have

$$\begin{aligned} [\hat{\phi}_{AB}, \hat{N}_{AB}] &\equiv [\hat{\phi}_A - \hat{\phi}_B, \hat{N}_A - \hat{N}_B] \\ &\approx -2i\mathbb{I}_{AB} + i d(|\Psi_A(\phi_0^A)\rangle\langle\Psi_A(\phi_0^A)| \otimes \mathbb{I}_B \\ &\quad + \mathbb{I}_A \otimes |\Psi_B(\phi_0^B)\rangle\langle\Psi_B(\phi_0^B)|). \end{aligned}$$

The Hilbert space of the system is now a tensor-product space spanned by states of the form $|\Psi(\phi_{AB})\rangle = |\Psi_A(\phi_A)\rangle \otimes |\Psi_B(\phi_B)\rangle$. With the reference phase difference defined as $\phi_0^{AB} \equiv \phi_0^A - \phi_0^B$, we obtain the expectation value:

$$\langle\Psi(\phi_{AB})|[\hat{\phi}_{AB}, \hat{N}_{AB}]|\Psi(\phi_{AB})\rangle \approx -2i(1 - d\delta_{\phi_0^{AB}, \phi_{AB}}). \quad (48)$$

III. PROJECTION OF AN LCJ CIRCUIT

Having derived an approximate projector onto the low-energy Hilbert space $\mathcal{H}_{\text{BCS}}$, we now derive the effective low-energy Hamiltonian for the circuit starting from the microscopic electronic Hamiltonian, as sketched in Fig. 1. We show that this approach gives results that are consistent with the conventional requantization of GL theory for systems with a large number of electrons.

We evaluate the projectors in Eq. (16) for an LCJ circuit up to second order in the electron tunneling through the junction barrier separating islands A and B . The lowest-order terms in the projection are

$$PH_0P = PH_AP + PH_BP + PH_{\text{Coul}}P. \quad (49)$$

Below, we evaluate the superconducting island terms H_A and H_B and the Coulomb interaction H_{Coul} , separately.

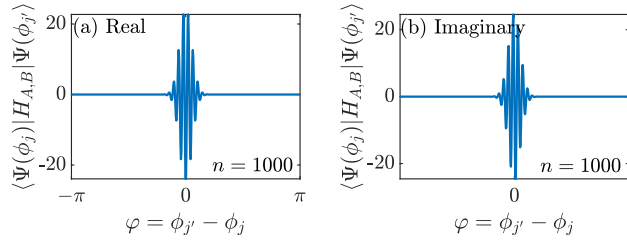


FIG. 5. Matrix elements $\langle \Psi(\phi_j) | H_{A,B} | \Psi(\phi_{j'}) \rangle$ for $n = 10^3$, $\Delta = 1$, and $b = 10$. The real and imaginary parts are illustrated in panels (a) and (b), respectively.

A. On-site energy: Intra-island terms

We compute the low-energy projections of the Hamiltonian term for island A from Eq. (1), with a similar expression for PH_BP . We find

$$\begin{aligned} PH_AP &= \sum_{jj'} \langle \Psi(\phi_j) | H_A | \Psi(\phi_{j'}) \rangle \langle \Psi(\phi_j) | \langle \Psi(\phi_{j'}) | \\ &= \sum_{jj'} (\mathcal{K}(\varphi_{jj'}) - |g|^2 \mathcal{V}(\varphi_{jj'})) |\Psi(\phi_j)\rangle \langle \Psi(\phi_{j'})|, \end{aligned} \quad (50)$$

where $\varphi_{jj'} = \phi_{j'} - \phi_j$ and

$$\mathcal{K}(\varphi) = 2 \sum_{k_A} \frac{(\epsilon_{k_A} - \mu_A) v_{k_A}^2}{u_{k_A}^2 + v_{k_A}^2} e^{i\varphi} \mathcal{W}(\varphi), \quad (51)$$

$$\mathcal{V}(\varphi) = \left(\sum_{k_A} \frac{u_{k_A} v_{k_A}}{u_{k_A}^2 + v_{k_A}^2} e^{i\varphi} \right)^2 e^{i\varphi} \mathcal{W}(\varphi). \quad (52)$$

We give a detailed calculation of the complex-valued functions $\mathcal{K}(\varphi)$ and $\mathcal{V}(\varphi)$ in Appendix B. Figure 5 shows the real and imaginary parts of the matrix element $\langle \Psi(\phi_j) | H_A | \Psi(\phi_{j'}) \rangle$, which, similar to $\mathcal{W}(\varphi)$, has a maximum at $\varphi = 0$ and tends to zero for $\varphi \neq 0$ and as $n \rightarrow \infty$. The diagonal terms in PH_AP give the mean ground-state energy:

$$\begin{aligned} \langle \Psi(\phi) | H_A | \Psi(\phi) \rangle &= \mathcal{K}(0) - |g_A|^2 \mathcal{V}(0) \\ &\approx -\frac{n_A \Delta_A^2}{2\mathcal{B}_A} \left(\ln \frac{\Delta_A}{2\mathcal{B}_A} + \lambda_A \ln \left(\frac{\Delta_A}{2\mathcal{B}_A} \right)^2 \right) - \mathcal{B}_A n_A / 2 \\ &\equiv E_A(\mathcal{B}_A, \Delta_A, \lambda_A) - E_F, \end{aligned} \quad (53)$$

where we have separated out the Fermi energy $E_F = \mathcal{B}_A n_A / 2$, and we have defined the electron-phonon coupling constant [22]:

$$\lambda_A = |g_A|^2 \varrho_F = |g_A|^2 n_A / (2\mathcal{B}_A), \quad (54)$$

where $\varrho_F = n_A / (2\mathcal{B}_A)$ is the noninteracting electronic density of states at the Fermi energy. The diagonal intra-island energy given in Eq. (53) is independent of ϕ_j , and corresponds to the “Mexican hat” potential. Interestingly, the on-site energy of an island, E_A , is not analytic at $\Delta_A = 0$, unlike the conventional quartic form used in GL theory.

We comment briefly that the off-diagonal terms that connect distinct values of ϕ_j and $\phi_{j'}$ in Eq. (50) are not strictly zero, but we discard them on the grounds that the overlap

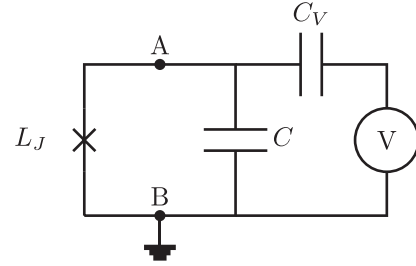


FIG. 6. A circuit consisting of a Josephson junction with a bias voltage V connected capacitively. The bias voltage creates an offset charge n_g to the capacitance Hamiltonian.

function tends quickly to zero. Were they to be retained, we speculate that they would result in an intrinsic, slow dispersion of initially localized phase states throughout the phase basis.

Minimum gap energy

The intra-island energy $E_A(\mathcal{B}_A, \Delta_A, \lambda_A)$ has a minimum value $E_A^{\min}$ at $\Delta_A^{\min}$, given at leading order in $1/\lambda_A$ by

$$\Delta_A^{\min} = 2\mathcal{B}_A e^{-1/\lambda_A}, \quad (55)$$

$$E_A^{\min} = -n_A \mathcal{B}_A e^{-2/\lambda_A}, \quad (56)$$

and similarly for island B. These values correspond to the standard GL MFT results [22].

For the majority of the following discussion, we will assume that the system relaxes to this energy minimum, and that there are no “Higgs”-like excitations away from this condition. We return briefly to this in Sec. IV B.

In the limit of large n , we have $\mathcal{W}(\phi_j - \phi_{j'}) \approx \delta_{jj'}$, so that Eq. (50) reduces to

$$PH_AP = E_A^{\min} P, \quad (57)$$

and similarly $PH_BP = E_B^{\min} P$. Thus, the first two terms of Eq. (49) are diagonal, constant energy offsets in the BCS subspace.

B. Capacitance: Interisland coulomb interaction

The last term in Eq. (49) is the interisland Coulomb interaction. Using the definition of H_{Coul} in Eq. (4) and the result of Eq. (41), we evaluate the low-energy projection of H_{Coul} in the number basis. For a fixed total charge $\hat{N}_A + \hat{N}_B = N_{\text{tot}}$, we have

$$\begin{aligned} PH_{\text{Coul}}P &= 4\lambda_C (P_A \hat{N}_A^{\text{CP}} P_A) (P_B \hat{N}_B^{\text{CP}} P_B) \\ &= 4\lambda_C \hat{N}_A \hat{N}_B \\ &= \lambda_C (N_{\text{tot}}^2 - \hat{N}_{AB}^2), \end{aligned} \quad (58)$$

where $\hat{N}_{AB} = \hat{N}_A - \hat{N}_B$ is the interisland charge difference. This is the usual quadratic capacitive coupling of two metallic islands.

Bias voltage: Briefly, a circuit may include a bias voltage as shown in Fig. 6. This shifts the equilibrium charge on each island as $N_A \rightarrow N_A + C_\Sigma V / (2e)$ and $N_B \rightarrow N_B - C_\Sigma V / (2e)$, where C_Σ is the total island capacitance. Then, Eq. (58)

becomes

$$PH_{\text{Coul}}P = \lambda_C (N_{\text{tot}}^2 - (\hat{N}_{AB} - n_g)^2), \quad (59)$$

where $n_g = C_\Sigma V/e$.

Lowest-order Terms

Combining the results in Eqs. (57) and (58), the lowest-order projected Hamiltonian terms give

$$PH_0P \approx (E_A^{\min} + E_B^{\min})P + 4\lambda_C \hat{N}_A \hat{N}_B, \quad (60)$$

which are the island and capacitive energies, respectively.

C. Josephson effect: Tunneling

The lowest-order tunneling term is given by H_T [30], which changes the parity of the electron number on the two sides of the junction. Electronic wave functions of different charge parity are orthogonal, so the matrix element vanishes: $\langle \Psi(\phi_j) | H_T | \Psi(\phi_{j'}) \rangle = 0$. It follows that the first-order projection of the linear tunneling term vanishes,

$$PH_TP = 0. \quad (61)$$

This can be verified by detailed calculation.

The second-order tunneling contribution in the low-energy BCS subspace is given by [30]

$$\begin{aligned} PH_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_TP \\ = \sum_{jj'} \langle \Psi(\phi_j) | H_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_T | \Psi(\phi_{j'}) \rangle \\ \times | \Psi(\phi_j) \rangle \langle \Psi(\phi_{j'}) |, \end{aligned} \quad (62)$$

where $E_0 = E_A + E_B + \lambda_C N_A^e N_B^e$ is the unperturbed energy of the metallic island. The diagonal matrix elements inside the sum (for $j = j'$) can be evaluated from the Fourier transform of the Green's function [38,39]:

$$\begin{aligned} \langle \Psi(\phi_j) | H_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_T | \Psi(\phi_j) \rangle \\ = \mathcal{F}_{\tau \rightarrow E_0} \langle \mathcal{T}_\tau [H_T(\tau)H_T] \rangle \\ = 2 \sum_{\mathbf{k}_A \mathbf{k}_B} \mathbf{t}^2 e^{i\phi_j} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, \tau) \mathcal{F}_{\downarrow\uparrow}(\mathbf{k}_B, \tau) + \text{c.c.}, \end{aligned} \quad (63)$$

where $\mathcal{F}_{\tau \rightarrow E_0}$ is the Fourier transform from time to energy, and we assume a uniform tunneling matrix element $\mathbf{t}_{\mathbf{k}_A \mathbf{k}_B} = \mathbf{t}$, and $\mathcal{F}_{\uparrow\downarrow}$ and $\mathcal{F}_{\downarrow\uparrow}$ are anomalous Green's functions in reciprocal space, defined by $\mathcal{F}_{\sigma\sigma'}(\mathbf{k}, \tau) \equiv -\langle \mathcal{T}_\tau c_{-\mathbf{k}\sigma}^\dagger(\tau) c_{\mathbf{k}\sigma'}^\dagger \rangle$.

To compute off-diagonal matrix elements, we also include a phase displacement operator D_φ , defined for a single island as

$$D_\varphi = e^{i\varphi \hat{N}^{\text{CP}}}, \quad (64)$$

which displaces phase states as $D_\varphi |\Psi(\phi)\rangle = |\Psi(\phi + \varphi)\rangle$.

With this definition, the off-diagonal matrix elements in Eq. (62) are given by

$$\begin{aligned} \langle \Psi(\phi_j) | H_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_T | \Psi(\phi_{j'}) \rangle \\ = \langle \Psi(\phi_j) | H_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_T D_{\varphi_{jj'}} | \Psi(\phi_{j'}) \rangle \\ = \langle H_T(\tau)H_T D_{\varphi_{jj'}} \rangle \end{aligned}$$

$$\begin{aligned} &= 2 \sum_{\mathbf{k}_A \mathbf{k}_B} \mathbf{t}^2 e^{i\phi_j} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, \tau) \mathcal{F}_{\downarrow\uparrow}(\mathbf{k}_B, -\tau) \\ &\quad \times \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle + \text{c.c.} \\ &= 2\mathbf{t}^2 e^{i\phi_j} \frac{2N_A^e N_B^e}{b^2 - 1} \mathcal{I}(E_0, \Delta) + \text{c.c.}, \end{aligned} \quad (65)$$

where

$$\mathcal{I}(E_0, \Delta) = \begin{cases} \frac{\pi^2}{4\Delta} \left(1 - \frac{E_0^2}{16\Delta^2}\right), & \text{for } E_0 \ll \Delta, \\ \frac{\pi \ln(2E_0/\Delta)}{2E_0}, & \text{for } E_0 \gg \Delta. \end{cases} \quad (66)$$

Computational details are given in Appendix D.

The unperturbed energy E_0 is roughly the Coulomb interaction between the two superconductors, $E_0 \sim \lambda_C = E_C$, where $E_C = (2e)^2/(2C)$. For a typical capacitance $C \sim 10$ fF of a Cooper-pair-box system, we find that $E_C \sim 10^{-5}$ eV [40]. In general, the superconductor gap is $\sim 10^{-3}$ eV, such that $E_0/\Delta \sim 10^{-2}$, so we obtain $\mathcal{I} \approx \pi^2/2\Delta$ and then the second-order tunneling contribution is

$$\begin{aligned} PH_T(E_0\bar{P} - \bar{P}H_0\bar{P})^{-1}H_TP \\ \approx \frac{\pi^2 N_A^e N_B^e}{b^2} \frac{\mathbf{t}^2}{\Delta} \sum_{j,j'} e^{i\phi_j} \underbrace{\langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle}_{\approx \delta_{jj'}} | \Psi(\phi_j) \rangle \langle \Psi(\phi_{j'}) | \\ + \text{H.c.} \\ \equiv E_J \cos \hat{\phi}, \end{aligned} \quad (67)$$

where we have defined $\cos \hat{\phi} = \sum_j \cos \phi_j | \Psi(\phi_j) \rangle \langle \Psi(\phi_j) |$ and

$$E_J = \mathbf{t}^2 2\pi^2 N_A^e N_B^e \Delta / \mathcal{B}^2. \quad (68)$$

Equation (67) is the expected functional form describing the effective Hamiltonian for the well-known Josephson junction. We compare the estimate for E_J to physical values of the Josephson energy in Sec. IV A 1. Importantly, since the junction is pointlike, the phase coordinate is compact, $\phi \in \mathcal{Z} \subset [-\pi, \pi)$.

We note that previous work by Martinis and Osborne [41] and Ambegaokar and Baratoff [26] reported similar calculations to compute E_J as diagonal elements of H_T . Our results presented here include both diagonal and (small) off-diagonal elements resulting from the projection, which explicitly depend on the Fermi bandwidth. We also note that the projective approach described here could naturally be extended to higher orders in the tunneling, to capture higher-order Josephson harmonics with Hamiltonian contributions of the form $\cos(m\phi)$ [42].

D. Inductance: Spatially varying phase torsion

Here, we analyze the inductance of a superconducting wire, again based on projecting out a low-energy Hilbert space. The inductive energy arises from two contributions: the kinetic energy of the electrons in motion and the potential energy in the magnetic field induced by the current. Both terms contribute to the effective nodal flux Φ at each point in a circuit, which is itself determined by the superconducting phase ϕ .

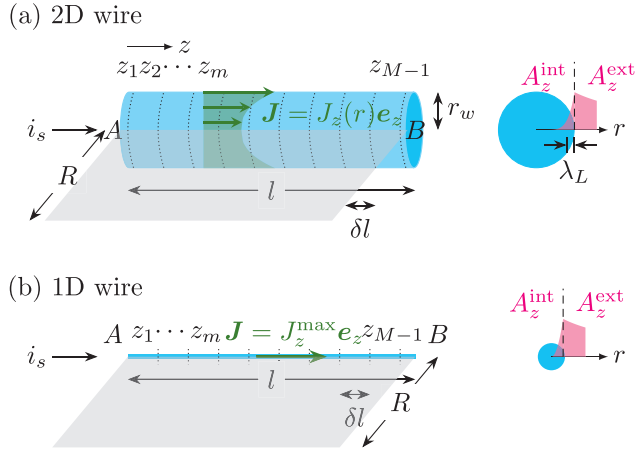


FIG. 7. (a) A cylindrically symmetric superconducting wire. The wire carries a supercurrent $i_s = \int_{\sigma} \mathbf{J} \cdot d\sigma$, which induces an internal field inside the superconductor (Appendix F). The external magnetic field penetrates a small depth $\lambda_L = \sqrt{m/(\mu_0 D e^2)}$ into the superconductor, where D is the electron density in the material and m is electron mass. (b) A 1D approximation to a wire. We consider the effective cross-section area $\sigma_{\text{eff}} = \pi r_w^2 - \pi(r_w - \lambda_L)^2 \approx 2\pi r_w \lambda_L$, i.e., the area that the applied vector potential can penetrate into. The supercurrent in the 1D model is the maximal value in the 2D model: $\mathbf{J} = J_z^{\text{max}} \mathbf{e}_z$.

We consider a superconducting wire with finite cross-section area shown in Fig. 7(a). For notational simplicity, we will take the wire to be cylindrical, so that the problem is a 2D system (while realistic microelectronic wires are 3D), and the superconducting phase and the vector potential depend on both the axial and radial coordinates, z and r . However, we seek to describe the problem as a quasi-1D wire, shown in Fig. 7(b), characterized by an axially varying phase $\phi(z)$.

Our goal is to derive the relevant Hilbert space and the constitutive relation for a lumped inductive element from the microscopic wave function of a distributed, spatially varying phase in a superconducting wire of length l . We start with a brief review of the classical analysis of a superconducting inductive wire, with more detail given in Appendix F.

Classically, the current density $\mathbf{J}(\mathbf{r})$ in a superconductor is related to the vector potential $\mathbf{A}^{\text{int}}(\mathbf{r})$ [43]:

$$\mathbf{J}(\mathbf{r}) = \frac{1}{\mu_0} \nabla \times \nabla \times \mathbf{A}^{\text{int}}(\mathbf{r}), \quad (69)$$

where $\mathbf{A}^{\text{int}}(\mathbf{r})$ is the vector potential on the interior of the wire and $\mathbf{J}(\mathbf{r}) = J_z(r) \mathbf{e}_z$ is axially symmetric. For convenience, we choose a gauge in which the vector potential is also axially symmetric, so that

$$\mathbf{A}(\mathbf{r}) = A_z(r) \mathbf{e}_z, \quad (70)$$

and we give more details about the form of $A_z(r)$ in Eq. (F6). In an axially symmetric gauge, we solve Eq. (69) to find

$$A_z^{\text{int}}(r) = -\mu_0 \int_0^r dr' r' \ln(r/r') J_z(r'). \quad (71)$$

That is, A_z^{int} is a linear functional of the current density.

More generally, the gauge field inside the superconductor is a linear functional of the current density:

$$\mathbf{A}^{\text{int}}(\mathbf{r}) = \mathbf{A}^{\text{int}}[\mathbf{J}], \quad (72)$$

which we use in subsequent analysis below. The total current in the wire is the integral of the current density over a cross section $\sigma = \pi r_w^2 \mathbf{e}_z$, i.e., $i_s \equiv \int_{\sigma} \mathbf{J} \cdot d\sigma$. Since the system is linear, the current density will scale linearly with total current in the wire $\mathbf{J}(\mathbf{r}) = i_s \mathbf{j}(\mathbf{r})$, where $\mathbf{j}$ is the normalized current density profile satisfying $\int_{\sigma} \mathbf{j} \cdot d\sigma = 1$, which carries information about the specific geometry of the wire. Putting these functional forms together, we have the general result:

$$\mathbf{A}^{\text{int}}[\mathbf{J}] = i_s \mathbf{A}^{\text{int}}[\mathbf{j}(r, z)], \quad (73)$$

indicating that the vector potential inside the superconductor scales linearly in the total current carried by the wire.

1. BCS subspace and projector for a 1D wire

We assume that a wire of finite diameter can be approximated by a quasi-1D wire shown in Fig. 7(b), which is characterized by an axially varying phase $\phi(z)$. Formally, we discretize the wire into $M \gg 1$ segments, so that the segmented Hilbert space of the wire is $\mathcal{H}_{\text{wire}} = \mathcal{H}_A \otimes \mathcal{H}_{z_1} \otimes \cdots \otimes \mathcal{H}_B$, in a basis spanned by states of the form

$$\begin{aligned} |\Psi(\phi(z))\rangle &= |\Psi_A(\phi_A)\rangle \otimes |\Psi_{z_1}(\phi_{z_1})\rangle \otimes \cdots \otimes |\Psi_B(\phi_B)\rangle \\ &\equiv \bigotimes_z |\Psi_z(\phi(z))\rangle, \end{aligned} \quad (74)$$

where z labels both the tensor-product element along the discretized wire and the phase coordinate at each position. In this formulation, there is a local compact Hilbert space associated with each segment.

Compared to the pointlike Josephson junction and capacitor model, the (distributed) inductive element has a significantly larger Hilbert space, which will become important below. We use the basis spanned by states of the form in Eq. (74) to define the low-energy projector over the length of the wire:

$$\begin{aligned} \Pi &= \sum_{\phi(z)} \bigotimes_z |\Psi_z(\phi(z))\rangle \langle \Psi_z(\phi(z))| \\ &= \bigotimes_z \sum_{\phi(z)} |\Psi_z(\phi(z))\rangle \langle \Psi_z(\phi(z))| \\ &= \bigotimes_z P_z, \end{aligned} \quad (75)$$

where $P_z = \sum_{\phi(z)} |\Psi_z(\phi(z))\rangle \langle \Psi_z(\phi(z))|$ defines a projector at each of the segments z along the wire, generalizing Eq. (15). Similarly, we define the phase-field and phase-gradient operators:

$$\hat{\phi}(z) = \bigotimes_z \sum_{\phi(z)} \phi(z) |\Psi_z(\phi(z))\rangle \langle \Psi_z(\phi(z))|, \quad (76)$$

$$\hat{\phi}'(z) = \bigotimes_z \sum_{\phi(z)} \phi'(z) |\Psi_z(\phi(z))\rangle \langle \Psi_z(\phi(z))|, \quad (77)$$

where the derivative in Eq. (77) is understood on the segmented wire as a finite difference. We note that the

summations in these field operators represent path-integral-like objects [19].

2. Inductive constitutive law

Using the second quantized form of the field operators $\psi_r = e^{i\phi(z)/2} c_r$ (where we suppress the spin label s), the (quantized) supercurrent density operator is given by [36,44]

$$\hat{\mathbf{J}}(\mathbf{r}) = \frac{e}{m} (\psi_r^\dagger (\hat{\mathbf{p}} - 2e\hat{\mathbf{A}}^{\text{int}}) \psi_r + \text{H.c.}), \quad (78)$$

where the quantized version of Eq. (72)

$$\hat{\mathbf{A}}^{\text{int}}(\mathbf{r}) = \mathbf{A}^{\text{int}}[\hat{\mathbf{J}}] \quad (79)$$

is itself a linear functional of the current density operator, which can be computed from Eq. (69) for any given geometry. Equation (78) therefore implicitly relates $\hat{\mathbf{J}}$ to the microscopic electronic degrees of freedom.

In the effective low-energy BCS subspace, the supercurrent density operator is given by the projection onto the low-energy subspace:

$$\begin{aligned} \hat{\mathbf{J}}^{\text{eff}}(\mathbf{r}, z) &= \Pi \hat{\mathbf{J}} \Pi = \bigotimes_z \sum_{\phi(z), \bar{\phi}(z)} \langle \Psi_z(\phi(z)) | \hat{\mathbf{J}}(\mathbf{r}) | \Psi_z(\bar{\phi}(z)) \rangle \\ &\quad \times | \Psi_z(\phi(z)) \rangle \langle \Psi_z(\bar{\phi}(z)) | \\ &\approx \bigotimes_z \sum_{\phi(z)} \langle \Psi_z(\phi(z)) | \hat{\mathbf{J}}(\mathbf{r}, z) | \Psi_z(\phi(z)) \rangle \\ &\quad \times | \Psi_z(\phi(z)) \rangle \langle \Psi_z(\phi(z)) | \\ &= \bigotimes_z \sum_{\phi(z)} \mathbf{J}(\mathbf{r}, z) | \Psi_z(\phi(z)) \rangle \langle \Psi_z(\phi(z)) |, \end{aligned} \quad (80)$$

where we have used the (approximate) orthonormality of the set $\{|\Psi(\phi(z))\rangle\}$ for each z . The current density matrix element is computed in Appendix E to be

$$\begin{aligned} \mathbf{J}(\mathbf{r}, z) &= \langle \Psi(\phi) | \hat{\mathbf{J}}(\mathbf{r}, z) | \Psi(\phi) \rangle \\ &= \frac{De^2}{m} \left(\frac{\Phi_0}{2\pi} \phi'(z) \mathbf{e}_z + \mathbf{A}^{\text{int}}(\mathbf{r}) \right), \end{aligned} \quad (81)$$

$$= \frac{De^2}{m} \left(\frac{\Phi_0}{2\pi} \phi'(z) \mathbf{e}_z + \mathbf{A}^{\text{int}}[\mathbf{J}] \right), \quad (82)$$

where m is the electron mass, e is the electron charge, and D is the electron density. The current-operator matrix element Eq. (81) is identical to the mean-field theory current density in a wire induced by gauge-invariant phase torsion [36]. Equation (82) makes clear that this expression represents an implicit definition of the projected current density matrix elements in terms of the BCS phase gradient ϕ' .

Putting together Eqs. (78)–(80) and (82), we find that the local current density operator in the low-energy BCS subspace is given by

$$\hat{\mathbf{J}}^{\text{eff}}(\mathbf{r}, z) = \frac{De^2}{m} \left(\frac{\Phi_0}{2\pi} \hat{\phi}'(z) \mathbf{e}_z + \mathbf{A}^{\text{int}}[\hat{\mathbf{J}}^{\text{eff}}] \right). \quad (83)$$

We are mainly interested in the mesoscopic properties of the wire, in particular, the net current integrated over the wire

cross section σ . Integrating both sides of Eq. (83) gives

$$\begin{aligned} \hat{i}_s(z) &= \int_\sigma \hat{\mathbf{J}}^{\text{eff}}(\mathbf{r}, z) \cdot d\sigma \\ &= \frac{De^2}{m} \left(\sigma_\times \frac{\Phi_0}{2\pi} \hat{\phi}'(z) + \hat{i}_s(z) \int_\sigma \mathbf{A}^{\text{int}}[\mathbf{J}(\mathbf{r}, z)] \cdot d\sigma \right), \end{aligned} \quad (84)$$

where $\sigma_\times$ is the cross-sectional area and we have used Eq. (73) in the second line. Solving for $\hat{i}_s(z)$ yields

$$\hat{i}_s(z) = \frac{\Phi_0}{2\pi} \hat{\phi}'(z) / \bar{L}, \quad (85)$$

where the total linear inductive density (i.e., inductance per unit length) is given by $\bar{L} = \bar{L}_K + \bar{L}_G$, which consists of the kinetic and geometric inductive densities:

$$\bar{L}_K = \frac{m}{De^2 \sigma_\times} \quad \text{and} \quad \bar{L}_G = -\frac{1}{\sigma_\times} \int_\sigma \mathbf{A}^{\text{int}}[\mathbf{J}(\mathbf{r}, z)] \cdot d\sigma. \quad (86)$$

For a concrete example, we consider a uniform cylindrical wire, in which we have an explicit form for the current density, given in Eq. (F2). In that case, the total integrated current operator is

$$\begin{aligned} \hat{i}_s(z) &= 2\pi \int_0^{r_w} dr r \hat{J}_z(r) \\ &= \frac{De^2}{2m} \left(\Phi_0 \hat{\phi}'(z) r_w^2 - \hat{i}_s(z) \mu_0 \lambda_L \left(2\lambda_L - \frac{r_w}{I_1(r_w/\lambda_L)} \right) \right), \end{aligned} \quad (87)$$

where $\lambda_L = \sqrt{m/(\mu_0 De^2)}$ is the London penetration depth. In the limit $\lambda_L \ll r_w$, $I_1(r_w/\lambda_L)^{-1}$ is exponentially suppressed. Solving for $\hat{i}_s(z)$ gives

$$\hat{i}_s(z) \approx \frac{\Phi_0}{2\pi} \hat{\phi}'(z) \frac{1}{\frac{m}{De^2} \frac{1}{\pi r_w^2} + \frac{\mu_0}{8\pi}} = \frac{\Phi_0}{2\pi} \hat{\phi}'(z) / \bar{L}, \quad (88)$$

where the total inductive density is given by $\bar{L} = \bar{L}_K + \bar{L}_G$, where $\bar{L}_G = \mu_0/(8\pi)$ [which agrees with the usual calculation of $\bar{L}_G$ up to a logarithmic factor in Eq. (F8)], and $\bar{L}_K$ is given in Eq. (86).

Finally, we make a lumped-element approximation, in which $\hat{\phi}'(z)$ is assumed to be uniform over the length of the wire, and is determined by the boundary values $\hat{\phi}_{A,B} \equiv \hat{\phi}(z_{A,B})$. That is, the phase-field operator simply interpolates linearly between the phases at each end of the wire $\hat{\phi}'(z) = (\hat{\phi}_B - \hat{\phi}_A)/l$, so that there is only a single effective current operator associated with the wire, given by

$$\hat{i}_s = (\hat{\phi}_B - \hat{\phi}_A)/L, \quad (89)$$

where $L = \bar{L}l$ is the total inductance, and

$$\hat{\phi}_{A,B} \equiv \frac{\Phi_0}{2\pi} \hat{\phi}_{A,B} \quad (90)$$

defines the lumped-element nodal fluxes associated with the end points of the wire. Equation (89) agrees with the usual constitutive inductive relation between current and flux. Importantly in Eq. (90), $\hat{\Phi}$ is derived from the underlying phase operator $\hat{\phi}$.

3. Effective Hamiltonian of an inductor

Here, we project the spatially varying fermionic Hamiltonian Eq. (1) onto the low-energy subspace $\mathcal{H}_{\text{BCS}}$, using a generalized projector Π that extends over the length of the wire. In the position basis in the presence of a vector potential, the electronic kinetic term in the island Hamiltonian H_A from Eq. (2) is given by

$$H_{K_A} = \int_{\text{SC}} dV c_{rs}^\dagger \left[\frac{1}{2m} (\mathbf{p} - e\mathbf{A}^{\text{int}})^2 - \mu \right] c_{rs}, \quad (91)$$

where the volumetric integral is over the interior of the superconducting metal. The effective Hamiltonian for an inductive wire is given by the projection

$$H_{\text{wire}} = \Pi (H_{K_A} + H_{I_A}) \Pi, \quad (92)$$

in which the electronic interaction term H_{I_A} contributes a constant $\mathcal{V}(\varphi)$, given in Eq. (52), which sets an energy offset in the inductive Hamiltonian.

To proceed, we remove the z dependence by introducing the gauge transformation $\mathcal{G}_{\phi(z)} = e^{i\phi(z)\hat{N}/2}$ and compute the projection:

$$\begin{aligned} H_{\text{wire}} &= \Pi H_{K_A} \Pi \\ &= \bigotimes_z \sum_{\phi(z), \bar{\phi}(z)} \langle \Psi_z(\phi(z)) | H_{K_A} | \Psi_z(\bar{\phi}(z)) \rangle \\ &\quad \times | \Psi_z(\phi(z)) \rangle \langle \Psi_z(\bar{\phi}(z)) | \\ &\approx \bigotimes_z \sum_{\phi(z)} \langle \Psi_z(\phi(z)) | H_{K_A} | \Psi_z(\phi(z)) \rangle \\ &\quad \times | \Psi_z(\phi(z)) \rangle \langle \Psi_z(\phi(z)) | \\ &= \frac{1}{2} \frac{De^2}{m} \int_{\text{SC}} dV \left(\frac{\Phi_0}{2\pi} \hat{\phi}'(z) - A_z^{\text{int}}(\hat{i}_s) \right)^2, \end{aligned}$$

where A_z^{int} is the vector potential inside the superconducting wire that couples to the current operator Eq. (88), “SC” denotes the interior volume of the superconductor, and the last line is a volumetric integral. Details of the derivation are given in Eqs. (E5)–(E10). For a superconducting wire with cylindrical symmetry, the vector potential operator is given by $A_z^{\text{int}}(\hat{i}_s) = \hat{i}_s \frac{\mu_0 \lambda_L}{2\pi r_w} \frac{1 - J_0(r/\lambda_L)}{J_1(r_w/\lambda_L)}$, computed in Eq. (F6). Expressing $\hat{i}_s$ in terms of $\hat{\phi}'$ using Eq. (88), we obtain

$$H_{\text{wire}} \approx \frac{1}{2} \left(\frac{\Phi_0}{2\pi} \right)^2 \frac{1}{L} \int_0^L dz \hat{\phi}'(z)^2, \quad \text{for } \lambda_L \ll r_w. \quad (93)$$

Details of the integration are given in Eq. (E13).

4. Lumped-element inductive energy

We evaluate the integral in Eq. (93) in the lumped-element approximation in which $\hat{\phi}'(z) = (\hat{\phi}_B - \hat{\phi}_A)/l$ is independent of z , so that $\int_0^L dz \hat{\phi}'^2 = (\hat{\phi}_B - \hat{\phi}_A)^2/l$ and Eq. (93) yields

$$\begin{aligned} H_{\text{wire}}^{\text{lumped}} &= \frac{1}{2L} \left(\frac{\Phi_0}{2\pi} \right)^2 (\hat{\phi}_B - \hat{\phi}_A)^2 = \frac{E_L}{2} (\hat{\phi}_B - \hat{\phi}_A)^2 \\ &\equiv H_L, \end{aligned} \quad (94)$$

where we define the inductive energy of the lumped element as $E_L \equiv (\Phi_0/2\pi)^2/L$.

To make the lumped-element approximation more explicit, we rederive the expression for H_L in the segmented model for the wire. As in Eq. (74), we divide the wire into M segments, so that the finite difference approximation for the local phase gradient is $\hat{\phi}'(z_m) = (\hat{\phi}_{m+1} - \hat{\phi}_m)M/\ell$, and then Eq. (93) becomes

$$H_{\text{wire}} = \frac{E_L}{2} M \sum_{m=0}^M (\hat{\phi}_{m+1} - \hat{\phi}_m)^2, \quad (95)$$

where $m=0$ and $m=M$ label the exterior segments at the ends of the wire, z_A and z_B , respectively. This Hamiltonian acts on the Hilbert space $\mathcal{H}_A \otimes \cdots \otimes \mathcal{H}_B$, which is a large tensor product of local BCS subspaces.

To make the lumped-element approximation, we assume that the phases at the interior segments $0 < m < M$ are constrained to minimize the energy of the wire, given the exterior phases $\hat{\phi}_A$ and $\hat{\phi}_B$. It is straightforward to show that the partial minimization of H_{wire} over the values of the interior phases yields $\hat{\phi}_{m+1} - \hat{\phi}_m = (\hat{\phi}_B - \hat{\phi}_A)/M$. Substituting this into Eq. (95) yields Eq. (94).

5. Emergent low-energy Hilbert space of an inductor

While we have described a pointlike superconducting element (e.g., a capacitor or tunnel junction) with a periodic coordinate $\phi \in [-\pi, \pi)$, which respects the 2π ϕ -translation symmetries of these devices, the inductive Hamiltonian Eq. (94) breaks ϕ -translation symmetry, and the Hilbert space of $\hat{\phi}_{AB}$ is substantially larger. Here, we provide a physically motivated argument that provides an estimate on the size of the physically relevant Hilbert space for an inductive element.

As discussed above, the discretized 1D inductor is formed from M segments, and recalling the definition in Eq. (22) of $\mathcal{Z} \subset [-\pi, \pi)$, the natural Hilbert space for the wire is

$$\mathcal{H}_{\text{wire}} = \mathcal{H}_A \otimes \cdots \otimes \mathcal{H}_B = \mathcal{Z}^{\otimes M-1}, \quad (96)$$

where the exponent $M-1$ indicates that the relevant internal degrees of freedom are the relative phases $\delta\phi(z_m) = \phi(z_{m+1}) - \phi(z_m) \in \mathcal{Z}$ between neighboring segments.

The subsequent “lumped-element” approximation treats the dynamics of the interior segments as being determined by the boundary values ϕ_A and ϕ_B , and so the accessible states within $\mathcal{H}_{\text{wire}}$ satisfy $\delta\phi(z_m) = (\phi_B - \phi_A)/M \in \mathcal{Z}$ for each segment z_m . It follows that

$$\phi_B - \phi_A \in M\mathcal{Z} \subset [-M\pi, M\pi), \quad (97)$$

which determines the effective, lumped-element Hilbert space of the discretized wire.

We next note that the number of segments, M , is bounded by the physical constraint that the supercurrent cannot exceed the critical current i_c of the wire. In particular, the total current in the wire is $|i_s| = \Phi_0 |\phi_B - \phi_A|/L \leq \Phi_0 M\pi/L < i_c$, and so

$$M \leq i_c L / (\pi \Phi_0) \equiv M_{\text{max}}. \quad (98)$$

An estimate for the critical current $i_c = \frac{\Delta}{\lambda_L(T)} \sqrt{\frac{\rho_F}{2\mu_0}}$ [22] indicates that the accessible Hilbert space for the lumped-element inductive wire depends on the physical properties of the wire. It follows that when L is large, $M_{\text{max}} \gg 1$, and so the lumped-element phase difference operator $\hat{\phi}_B - \hat{\phi}_A$ has

support over a large but finite interval given by Eq. (97), which is bounded above by Eq. (98). Unlike the Hilbert space of the pointlike Josephson junction, the Hilbert space for the inductive element is no longer periodic, since lumped-element states with $\phi_B - \phi_A = \pm\pi M_{\max}$ correspond to orthogonal states with opposite current flows.

We comment in passing that this argument really asserts that the dissipationless, lumped-element effective model for an inductive element requires that the phase drop across the inductor should be bounded in the interval $\phi_B - \phi_A \in [-M_{\max}\pi, M_{\max}\pi]$. If $\phi_B - \phi_A$ were to exceed this range, the supercurrent would exceed the critical current, and dissipative quasiparticle formation would ensue.

A more formal treatment of this dissipative limit on the accessible Hilbert space for a superconducting inductor would require the inclusion of incoherent phase slips [36] that generate quasiparticles, which would require the expansion of the projective Hilbert space to include quasiparticle states, which lead to energy decay, charge fluctuations, and decoherence [11, 45–48]. We speculate that the dissipation would be described perturbatively by open-system Lindblad superoperators whose effect would be to suppress large deviations in $\phi_B - \phi_A$. This is beyond the goals of this work, and we leave this to future research.

6. Offset-charge dependence in superconducting devices

We comment here on the *dependence* of a pointlike device (e.g., a transmon) on the charge offset n_g , compared to the *independence* of a spatially extended inductive device (e.g., fluxonium) on n_g , as noted in Ref. [49]. Addressing this, Thanh Le *et al.* [17] provided a comparison of the discrete, charge-offset-dependent spectrum $\mathbf{S}_{\text{compact}}(n_g)$, arising from a compact description of the transmon phase operator ($-\pi \leq \phi < \pi$), compared to the continuum Bloch band structure $\mathbf{S}_{\text{ext}} = \bigcup_{n_g} \mathbf{S}_{\text{compact}}(n_g)$, arising from an extended (noncompact) description of the phase. Since $\mathbf{S}_{\text{ext}}$ is itself the union of $\mathbf{S}_{\text{compact}}(n_g)$ over all values of n_g , it is consequently independent of n_g . In this respect, the *kinematic* structure of the eigenspace of fluxonium and the extended-phase description of a transmon are both independent of n_g .

However, a further conclusion of Ref. [17] was that for the extended-phase description of the periodic transmon Hamiltonian, the quasicharge is an exactly conserved quantum number, and so only direct, interband transitions are allowed—unless a (hypothetical) switchable inductive element is introduced to the circuit. The periodic translation symmetry ensures that indirect transitions (which change the quasicharge) are forbidden, so that in a given experiment, only a discrete subspace $\mathbf{S}_{\text{compact}}(n_g) \subset \mathbf{S}_{\text{ext}}$ is dynamically accessible, for any fixed n_g . In this respect, the *dynamics* of the compact-phase and extended-phase descriptions of a transmon will both depend on n_g , as is commonly seen experimentally. This is in contrast to fluxonium in which the quasicharge is not conserved by symmetry, and for which the spectrum and dynamics are both independent of n_g .

Table I summarizes these classifications: Fluxonium or other inductive devices are kinematically and dynamically independent of n_g (bottom-right cell, purple), while a compact-phase transmon is kinematically and dynamically dependent on n_g (top-left cell, green). A hypothetical extended-phase

TABLE I. Summary of *kinematic* (“kin.”) and *dynamic* (“dyn.”) dependence or independence of different systems on the charge offset n_g , determined by the compact/extended phase support, and the periodicity of the Hamiltonian. Transmons and other pointlike devices are in the top-left box (green), while spatially extended inductive elements are in the bottom-right box (purple).

Dependence on charge offset		Phase Support	
		Compact	Extended
Hamiltonian Type	Periodic	kin. depend dyn. depend	kin. indep dyn. depend
	Aperiodic	-	kin. indep dyn. indep

transmon, described in Ref. [17] as the limit of a transmon with an infinitesimally small inductive shunt, would exhibit a mixture of these properties: kinematically independent and dynamically dependent (top-right cell, shaded). Kinematics determines the structure of the energy eigenspace, whereas dynamics determines the experimental (i.e., dynamical) response.

The additional insight we add here is that the phase coordinate for a pointlike transmon (or Cooper-pair box) is genuinely compact, arising from the functional periodicity of the BCS wave function for a pointlike superconductor or junction $|\Psi(\phi + 2\pi)\rangle = |\Psi(\phi)\rangle$, whereas that of the spatially extended inductive element in fluxonium requires an extended description of the phase operator, as discussed in Sec. III D 5. In this respect, the dependence, or otherwise, of the spectrum (and the dynamics) on n_g , as discussed in Ref. [49], is a consequence of the compactness of the transmon phase, in contrast to the extended-phase coordinate of fluxonium or other spatially extended inductive devices.

E. Effective LCJ Hamiltonian

Combining the results given above, the effective Hamiltonian for the LCJ circuit shown in Fig. 2 is

$$H_{\text{LCJ}} \approx E_C \hat{N}_{AB}^2 - E_J \cos \hat{\phi}_{AB} + \frac{E_L}{2} \hat{\phi}_{AB}^2, \quad (99)$$

where $\hat{N}_{AB} = \hat{N}_B - \hat{N}_A$ and $\hat{\phi}_{AB} = \hat{\phi}_B - \hat{\phi}_A$. This agrees with the standard expression for a quantized LCJ circuit derived by requantizing GL mean-field theory. However, the microscopic derivation that we have undertaken here provides clarity on the low-energy Hilbert space for each component.

Different choices of the energy scales among the energy coefficients result in various potential landscapes for well-known superconducting qubits [50].

IV. DISCUSSION

A. Effective capacitor-junction Hamiltonian

Transmon and Cooper-pair box qubits are formed from a capacitor and a Josephson junction, so Eq. (99) reduces to

$$\begin{aligned} H_{\text{CJ}} &= E_C (\hat{N} - n_g)^2 - E_J \cos \hat{\phi} \\ &= E_C (\hat{N} - n_g)^2 - E_J (e^{i\hat{\phi}} + e^{-i\hat{\phi}})/2, \end{aligned} \quad (100)$$

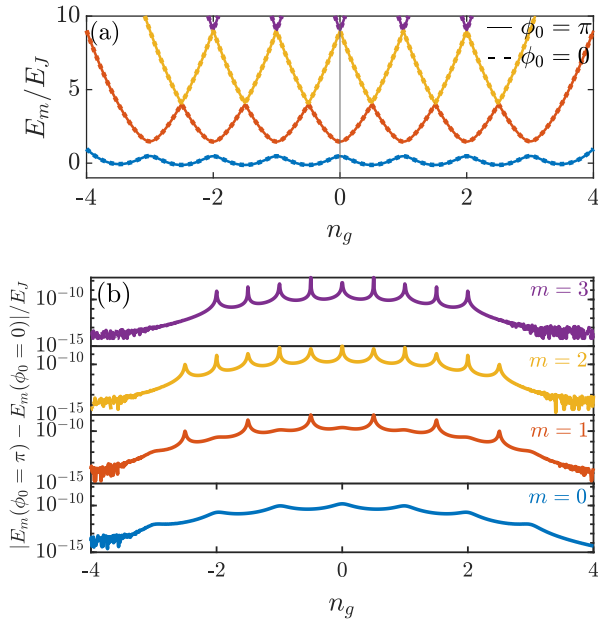


FIG. 8. (a) The eigenenergies E_m for the first four levels of a transmon in Eq. (100) for $d = 8$ with $E_J = E_C$. The solid thin lines are with the reference phase $\phi_0 = \pi$, while the dashed lines are with $\phi_0 = 0$. The maximal energy difference between each solid and dashed is of the order $\mathcal{O}10^{-10}$ and thus can be viewed as overlap. All energies are scaled by E_J . (b) Energy differences between two choices of reference phase ($\phi_0 = 0$ and π) for $m = 0, 1, 2, 3$ levels.

where the first term corresponds to Eq. (58) and the second term corresponds to Eq. (67), in which the Cooper-pair ladder operator is [29]

$$e^{i\hat{\phi}} = \sum_{j=1}^{d-1} |\tilde{\Psi}(j)\rangle \langle \tilde{\Psi}(j+1)| + e^{id\phi_0} |\tilde{\Psi}(d)\rangle \langle \tilde{\Psi}(1)|. \quad (101)$$

Since the transmon circuit is spatially pointlike, with no extended inductive elements, the phase operator $\hat{\phi}$ is compact on the interval $\phi \in [-\pi, \pi)$, and the number operator is correspondingly discrete. Importantly, there is a residual dependence on ϕ_0 , which arises from the compactness of the Hilbert space [29]. We address here the effect of this parameter choice. Figure 8(a) shows the calculated eigenspectra of Eq. (100) as a function of the gate bias n_g , for $\phi_0 = 0$ (dashed) and π (solid), which are almost identical, as shown by the difference plotted in Fig. 8(b). These plots were generated for the unphysically small value $d = 8$, and the difference decreases as d increases. Practically, this means that the value of ϕ_0 is insignificant for realistic systems.

In Fig. 9, we compute the effect of ϕ_0 on the spectra of transmons for which $E_J/E_C \sim 10^2$ [51] and phase qubits for which $E_J/E_C \sim 10^4$ [52]. In these systems, we have used $d = 18$ for the transmon case, and $d = 48$ for the phase qubit case, both of which are smaller than would be expected in a real system, indicating that these systems would be practically insensitive to ϕ_0 as well.

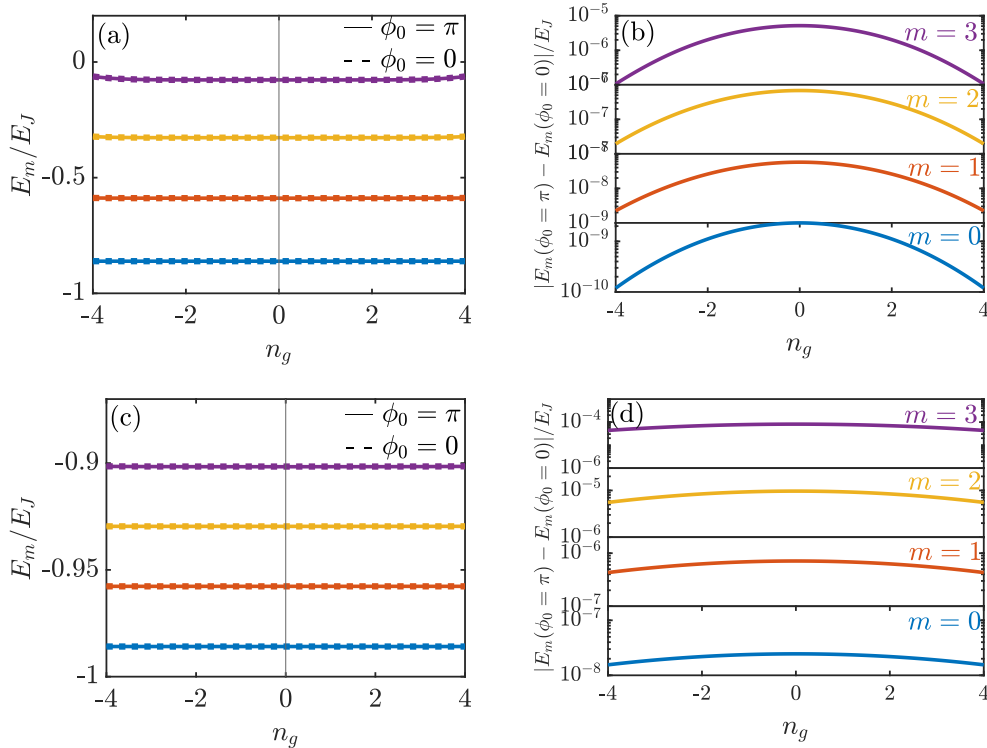


FIG. 9. The eigenenergies and the energy differences for a transmon with $E_J/E_C \sim 10^2$ [(a) and (b)] and a phase qubit with $E_J/E_C \sim 10^4$ [(c) and (d)]. The solid and dotted lines in panels (a) and (c) represent the results for $\phi_0 = \pi$ and $\phi_0 = 0$, respectively.

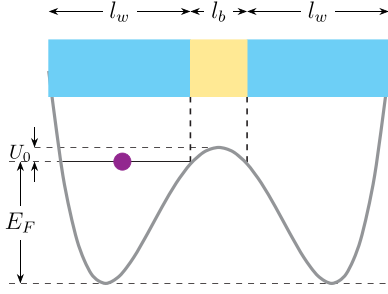


FIG. 10. We estimate the tunnel matrix element t using the WKB approximation in a double-well potential. The oxide layer acts as the barrier, whose energy is U_0 —higher than the electron energy inside the superconductors.

1. Experimental parameter estimates

Here, we compare the microscopic expression for the junction energy in Eq. (67) with physically measured values. Typical aluminum parameters are $\Delta \approx 340 \mu\text{eV}$ [53], electronic bandwidth $B \approx 10.6 \text{ eV}$ [54], so $b \approx 3 \times 10^4$. We adopt device parameters in Ref. [55], and assume the metallic islands forming the junction are identical to one another: The thickness of the aluminum islands and the oxide barrier are approximately $l_w \approx 30 \text{ nm}$ and $l_b \approx 1 \text{ nm}$, respectively, the area of the junction is $A_J \approx 0.02 \mu\text{m}^2$, and the measured junction energy is $E_J^{[56]} \approx 270 \mu\text{eV}$. With this geometry, and for a face-centered cubic aluminum unit cell of side length $l_c = 0.4 \text{ nm}$ that contains four atoms per unit cell, each metallic island contains approximately $N_{A,B}^e = N_e = 4l_w A_J / l_c^3 \approx 3 \times 10^7$ electrons.

Microscopic tunneling from the WKB (Wentzel-Kramers-Brillouin) method: We model the junction barrier as a one-dimensional double-well potential, shown in Fig. 10. Electrons within each well travel with Fermi velocity $v_F \approx 2 \times 10^6 \text{ m/s}$ for aluminum [56], colliding with the barrier with an “attempt” frequency $f = v_F / (2l_w) \approx 30 \text{ THz}$.

According to the WKB method, the tunnel matrix element t is given by [57]

$$t = h f T_{\text{WKB}}, \quad (102)$$

where

$$T_{\text{WKB}} = e^{-2\xi l_b \sqrt{2m_e^* U_0 / \hbar}} \quad (103)$$

is the WKB tunneling amplitude per collision. In this expression, $U_0 \sim 2 \text{ eV}$ is the barrier height [58,59], the effective electron mass is $m_e^* = 0.4m_e$ for aluminum oxide [60], and we include a dimensionless “fudge factor” $\xi \sim 1$ to account for relative uncertainties in the dimensional factors appearing in the exponent of T_{WKB} .

Assuming symmetric metal islands and adopting the geometric parameters in Ref. [55], the estimated junction energy is given by Eq. (68), $E_J \approx t^2 2\pi^2 N_A^e N_B^e \Delta / B^2$, which depends exponentially on ξ . Figure 11 shows the value of E_J estimated from Eq. (68), and for $\xi \approx 1.65$, we see that E_J matches the observed value $E_J^{[56]}$. This indicates that Eq. (68) plausibly defines E_J .

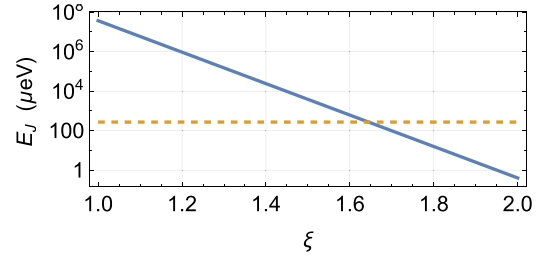


FIG. 11. The estimated junction energy E_J , as a function of the “fudge factor” ξ in the tunneling amplitude, given by Eqs. (68) and (103) (solid line). The experimentally measured value of $E_J^{[56]} \approx 270 \mu\text{eV}$ is also shown (dashed line) and matches the derived value of E_J at $\xi \approx 1.6$.

2. WKB in nonzero magnetic fields

Consider a Josephson junction in an external vector potential $\mathbf{A}$. The tunnel matrix element is oscillatory in phase [61]:

$$t' = t e^{i \int_{\Gamma} \mathbf{A} \cdot d\mathbf{l} / \Phi_0} = t e^{i \Phi_b l_b / (2l_w \Phi_0)}, \quad (104)$$

where t is the regular tunnel matrix element without fields, $\mathbf{l}$ is the tunneling path, and Φ_b is the bias magnetic flux threading the device. The tunneling Hamiltonian gains a phase shift in t , becoming

$$\begin{aligned} E_J \cos \hat{\phi} &\mapsto E_J \cos \left(\hat{\phi} - \frac{l_b}{2l_w} \Phi_b / \Phi_0 \right) \\ &\approx E_J \cos \hat{\phi}, \end{aligned} \quad (105)$$

where the second line assumes the insulator is much thinner than the well width, $l_w \gg l_b$. We see that the bias offset ordinarily would not manifest itself in the constitutive relation for a pointlike junction, and should instead be accounted for extended inductive loops [62].

B. Higgs modes

In this work, we have treated the phase $\phi = \arg(\Delta)$ as a “position” coordinate, and used it to define a basis for the BCS subspace. From this basis, we constructed the low-energy phase operator $\hat{\phi}$ in Eq. (39), and via the Fourier transform, its conjugate $\hat{n}$. In the foregoing analysis, we have treated $|\Delta|$ as a constant, determined by minimizing the ground-space energy of a metallic island, given by Eq. (53). Here, we briefly comment on some further considerations of this energy landscape.

The ground-state energy of a superconducting island given in Eq. (53) can be conveniently rescaled relative to its minimum value $E^{\min} = -B e^{-2/\lambda} n$ and the location of the energetic minimum at $\Delta^{\min} = 2B e^{-1/\lambda}$. We define the dimensionless gap coordinate $d \equiv \Delta / \Delta^{\min}$, so the rescaled version of Eq. (53) is

$$\begin{aligned} \mathcal{E} &\equiv E / |E^{\min}| \\ &\approx d^2 (-1 + 2 \ln d - 2\lambda (\ln d)^2) \\ &\approx -1 + 2(1 - \lambda)(d - 1)^2 + O(d - 1)^3, \end{aligned} \quad (106)$$

where the last line is a quadratic approximation for the energy around the minimum at $d = 1$. Figure 12(a) shows $\mathcal{E}$, which

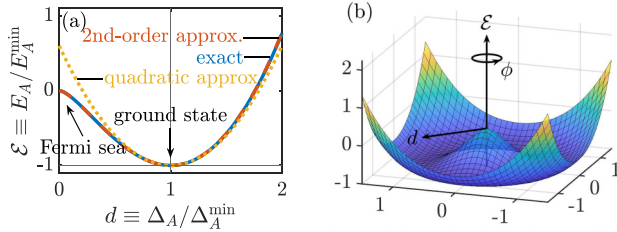


FIG. 12. (a) The scaled energy spectrum $E_A/E_A^{\min}$ with respect to the scaled order parameter $d \equiv \Delta_A/\Delta_A^{\min}$. The second-order approximation in Eq. (53) (orange dashed) matches closely with the exact solution (blue solid) given in Eq. (B4), for large values of the dimensionless bandwidth b . At $\Delta_A = \Delta_A^{\min}$, we obtain the minimum of the energy spectrum that corresponds to the gap parameter at zero temperature. We use the quadratic expansion in Eq. (106) (dotted yellow) to estimate the effective mass of the quasiparticle from the harmonic oscillator model. (b) The Mexican-hat-shaped potential well of the superconductor with respect to the values of ϕ , represented as an azimuthal angle, and d as the radial coordinate.

corresponds to the radial coordinate of the Mexican-hat potential.

For an isolated metallic island, $\mathcal{E}$ is independent of ϕ , so $\mathcal{E}$ is symmetric in rotations of ϕ . Plotting $\mathcal{E}$ as a function of the dimensionless gap coordinate d (radial) and phase angle ϕ (azimuthal) yields the Mexican-hat potential, shown in Fig. 12(b). In this potential, the low-energy dynamics discussed in earlier sections are restricted to the minimum of this potential, and we have hitherto assumed that $\Delta = \Delta^{\min}$.

However, a real system can also undergo “Higgs”-like excitations of the radial coordinate. Quantum mechanically there may be excitations of the Higgs mode, and so superconducting devices should more generally be analyzed over a two-dimensional polar coordinate space (ϕ, Δ_r) , where ϕ is the azimuthal angle of the Mexican hat and Δ_r is the radial gap coordinate. The phaselike mode is massless, while the radial-like gap mode (or Higgs mode) has some effective massive.

We briefly comment on the structure of this larger coordinate space. We suppose that the BCS ground states are now a function of both ϕ and Δ_r , so that we would generalize the basis defined in Eq. (12) to a larger space spanned by the set of states $\{|\Psi(\phi_j, \Delta_r)\rangle\}_{j,r}$, where Δ_r is now part of the parametrization of the larger basis.

As in Sec. II, we should analyze the overlap function $\mathcal{Y}(\phi_j, \phi_{j'}; \Delta_r, \Delta_{r'}) = \langle \Psi(\phi_j, \Delta_r) | \Psi(\phi_{j'}, \Delta_{r'}) \rangle$ in order to understand the effective dimension of this larger space. As a start in this direction, for $j = j'$, we compute

$$\begin{aligned} \mathcal{Y}(\phi_j, \phi_j; \Delta_r, \Delta_{r'}) &= \langle \Psi(\phi_j, \Delta_r) | \Psi(\phi_j, \Delta_{r'}) \rangle \\ &\approx e^{-\frac{3n\pi(\Delta_{r'} - \Delta_r)^2}{64\sqrt{2b}\Delta_r^2}}, \end{aligned} \quad (107)$$

where the second line is evaluated using a similar procedure as used to compute $\mathcal{W}(\varphi)$, given in Eq. (21). $\mathcal{Y}$ is approximately a Gaussian with width $\delta\Delta \sim \sqrt{B\Delta/n}$, and so we would expect that a discrete subset of radial modes would describe the space with high fidelity, similar to the emergent discrete set of phase states, described earlier. Namely, the number of nearly

orthogonal transverse modes that can be accommodated in the potential well $E_A(\mathcal{B}, \Delta)$ for a given phase ϕ is roughly $\Delta^{\min}/\delta\Delta \sim \sqrt{ne^{-1/\lambda}}$. We leave more extensive analysis to future work.

The Higgs mode of a superconductor can be excited resonantly by a driving field with energy at $\Delta^{\min}$, and oscillates at frequency $\omega_{\text{Higgs}} = 2\Delta^{\min}/\hbar$ [10]. By treating the lowest-energy Higgs mode as a harmonic oscillator with the quadratic potential in Eq. (106), we obtain the effective mass of the oscillator associated with the potential $E(\mathcal{B}, \Delta)$:

$$m^* \approx \frac{\hbar^2}{4} \cdot \frac{1}{\rho^2(E_F)|E^{\min}|}. \quad (108)$$

The depth of the Higgs potential in Fig. 12 can be expressed in terms of ω_{Higgs} , and we estimate the number of bound Higgs-like excitations to be

$$\# \text{ bound states} = |E^{\min}|/(\hbar \omega_{\text{Higgs}}) \sim ne^{-1/\lambda}. \quad (109)$$

There is thus a nontrivial space of bound Higgs excitations that may be accessible in superconducting quantum devices, and these may be useful or interesting for quantum technologies. We leave this to further work.

V. CONCLUSION

We have described here a direct approach to the quantization of superconducting circuits. By projecting the microscopic Hamiltonian onto the low-energy BCS subspace, we derive the effective Hamiltonian of a superconducting LCJ circuit from first principles.

With this fully quantum framework, we establish a direct pathway from the microscopic description to the macroscopic picture of a quantized circuit. The low-energy BCS space has a natural discretization that emerges from the structure of the BCS overlap function. We have shown how to construct phase and charge operators on this subspace, and derived their commutation relation.

The projective approach we have described here reproduces the known results for capacitive, inductive, and Josephson tunneling elements. One conclusion from our analysis is that since inductors are inherently extended objects, the Hilbert space associated with them is larger than that for pointlike capacitive and Josephson tunneling elements, whose Hilbert space is periodic on the interval $[-\pi, \pi)$. Although Devoret [63] has argued the support of the gauge-invariant superconducting phase (i.e., whether it is a “line” or a “circle”), our approach resolves some of the questions about the compactness or otherwise of superconducting circuits reviewed in Ref. [17] based on a microscopic derivation from first principles.

The approach we take here opens a number of directions for further work. First, we have given a physically motivated argument to estimate the support of the accessible Hilbert space of an inductive element, and we postulate that this will emerge dynamically from decoherence induced by quasiparticle formation at sufficiently high currents. This could be formalized by extending the projective subspace to include few quasiparticle states. These additional modes will produce decoherence relative to the BCS subspace, and should

dynamically constrain the accessible states of extended wire-like inductive elements.

Second, while there has been progress since Widom *et al.* [13] in developing the theory of quantum-phase slip devices (e.g., see Ref. [64]), to our knowledge there is as yet no microscopic Hamiltonian derivation analogous to that for Josephson junctions. We expect that a detailed projective analysis of superconducting wires whose thickness is comparable to or smaller than the penetration depth provides a viable pathway to realize such a derivation.

Lastly, while we have focused on the superconducting phase degree of freedom ϕ , the BCS family of states admits the possibility of quantizing the gap amplitude $|\Delta|$, which should yield Higgs-like excitations of superconducting islands. We have provided a starting point for this analysis, and speculate that this may provide for a greatly expanded family of superconducting quantum devices that could be built experimentally.

ACKNOWLEDGMENTS

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DATA AVAILABILITY

No data were created or analyzed in this study.

APPENDIX A: EIGENSTATE OF ELECTRON NUMBER OPERATOR

In this Appendix, we show that the number state defined in Eq. (40) is the eigenstate of both electron and Cooper-pair number operators, $\hat{N}^{\text{el}}$ and $\hat{N}$, respectively.

The BCS state in N representation is described by

$$|\tilde{\Psi}(N)\rangle = \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \right)^{1/2} \times \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{-iN\phi_j} \prod_{k=k_1}^{k_n} (u_k + v_k e^{i\phi_j} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle. \quad (\text{A1})$$

We can expand the product of binomials in Eq. (A1) into a sum of product by the Lemma [65]:

Lemma A1. $\forall A_k, B_k$ such that $[A_k, B_k] = 0$, we have an identity

$$\prod_{k=1}^n (A_k + B_k) = \sum_{m=0}^n \sum_{\chi \in \mathcal{S}_m} \prod_{i \in \chi^c} A_i \prod_{j \in \chi} B_j,$$

where $\mathcal{S}_m$ is the set of all subsets of $\{1, 2, \dots, n\}$ with exactly m elements, i.e.,

$$\mathcal{S}_m \equiv \{\chi \subseteq \{1, 2, \dots, n\}, |\chi| = m, \}, \quad |\mathcal{S}_m| = \binom{n}{m},$$

and χ^c is the complement of χ .

With the substitution $u_k \rightarrow A_k$, $v_k e^{i\phi_j} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger \rightarrow B_k$, Eq. (A1) becomes

$$|\tilde{\Psi}(N)\rangle = \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \right)^{1/2} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{-iN\phi_j} \sum_{m=0}^n \sum_{\chi \in \mathcal{S}_m} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} e^{i\phi_j} c_{k_j\uparrow}^\dagger c_{-k_j\downarrow}^\dagger |0\rangle, \quad (\text{A2a})$$

$$\begin{aligned} &= \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \right)^{-1/2} \frac{\alpha}{\sqrt{n-1} + \alpha} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{-iN\phi_j} \sum_{m=0}^n e^{im\phi_j} \sum_{\chi \in \mathcal{S}_m} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j\uparrow}^\dagger c_{-k_j\downarrow}^\dagger |0\rangle \\ &= \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \right)^{-1/2} \sum_{m=0}^n \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{i(m-N)\phi_j} \right) \sum_{\chi \in \mathcal{S}_m} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j\uparrow}^\dagger c_{-k_j\downarrow}^\dagger |0\rangle. \end{aligned} \quad (\text{A2b})$$

For large n such that $\sqrt{n} \gg \alpha$, we can define a new variable $n' = \sqrt{n}/\alpha \approx \sqrt{n-1}/\alpha$, so the term inside the brackets of Eq. (A2b) can be rewritten as

$$\begin{aligned} \frac{\alpha}{\sqrt{n-1} + \alpha} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{i(m-N)\phi_j} &= \frac{\alpha}{\sqrt{n-1} + \alpha} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{i(m-N)(-\pi + 2\pi\alpha j/\sqrt{n})} \\ &\approx \frac{1}{n' + 1} \sum_{j=0}^{n'} e^{i(m-N)(-\pi + 2\pi j/n')} = e^{-\pi i(m-N)} \cdot \frac{1}{n' + 1} \sum_{j=0}^{n'} e^{2\pi i(m-N)j/n'}. \end{aligned}$$

Taking the limit $n \rightarrow \infty$ and $n' \rightarrow \infty$, we have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\alpha}{\sqrt{n-1} + \alpha} \sum_{j=0}^{\sqrt{n-1}/\alpha} e^{i(m-N)\phi_j} &= e^{-\pi i(m-N)} \lim_{n' \rightarrow \infty} \frac{1}{n' + 1} \sum_{j=0}^{n'} e^{2\pi i(m-N)j/n'} \\ &\approx e^{-\pi i(m-N)} \cdot \underbrace{\lim_{n' \rightarrow \infty} \frac{1}{n'} \sum_{j=0}^{n'} e^{2\pi i(m-N)j/n'} }_{=\delta_{mN}} = e^{-\pi i(m-N)} \delta_{m,N}, \end{aligned}$$

where $\delta_{m,N}$ is the Kronecker delta function. Now Eq. (A2b) is

$$\begin{aligned} |\tilde{\Psi}(N)\rangle &= \left(\frac{\alpha}{\sqrt{n-1} + \alpha} \right)^{-1/2} \sum_{m=0}^n e^{-\pi i(m-N)} \delta_{m,N} \sum_{\chi \in \mathcal{S}_m} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle \\ &= \frac{1}{\sqrt{d}} \sum_{\chi \in \mathcal{S}_N} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle. \end{aligned} \quad (\text{A3})$$

Next, with Eq. (A3), the electron number operator $\hat{N}^{\text{el}}$ of a single superconductor acting on this number state $|\tilde{\Psi}(N)\rangle$ gives

$$\begin{aligned} \hat{N}^{\text{el}} |\tilde{\Psi}(N)\rangle &= \sum_{k'} (c_{k'\uparrow}^\dagger c_{k'\uparrow} + c_{-k'\downarrow}^\dagger c_{-k'\downarrow}) \frac{1}{\sqrt{d}} \sum_{\chi \in \mathcal{S}_N} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle \\ &= \frac{1}{\sqrt{d}} \sum_{\chi \in \mathcal{S}_N} \sum_{k' \in \chi} \prod_{k_i \in \chi^c} u_{k_i} (c_{k'\uparrow}^\dagger c_{k'\uparrow} + c_{-k'\downarrow}^\dagger c_{-k'\downarrow}) \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle \\ &= \frac{1}{\sqrt{d}} \sum_{\chi \in \mathcal{S}_N} \sum_{k' \in \chi} 2 \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle \\ &= 2N \left(\frac{1}{\sqrt{d}} \sum_{\chi \in \mathcal{S}_N} \prod_{k_i \in \chi^c} u_{k_i} \prod_{k_j \in \chi} v_{k_j} c_{k_j}^\dagger c_{-k_j}^\dagger |0\rangle \right) \quad \because |\chi| = N \\ &= 2N |\tilde{\Psi}(N)\rangle, \end{aligned} \quad (\text{A4})$$

where χ is a subset of the single-particle modes. The eigenvalue equation $\hat{N}^{\text{el}} |\tilde{\Psi}(N)\rangle = 2N |\tilde{\Psi}(N)\rangle$ implies that we have exactly $2N$ electrons in the state $|\tilde{\Psi}(N)\rangle$.

APPENDIX B: MATRIX ELEMENTS OF THE SUPERCONDUCTING ISLAND HAMILTONIAN

The matrix elements of the island Hamiltonian projected onto the low-energy Hilbert space in Eq. (50) are given by three momentum sums, which can be approximated by the integrals: $\sum_{k=-\infty}^{\infty} \rightarrow 2 \int_{-\Delta}^{\Delta} dE \rho(E)$, where $\rho(E) = \frac{n}{2\Delta\sqrt{b^2-1}} \frac{E}{\sqrt{E^2-\Delta^2}}$. Recalling u_k and v_k from Eq. (11) and using the results

$$\begin{aligned} \langle \Psi(\phi_j) | c_{k\uparrow}^\dagger c_{k\uparrow} | \Psi(\phi_{j'}) \rangle &= \langle 0 | \prod_{k' \neq k} (u_{k'} + v_{k'} e^{-i\phi_j} c_{-k'\downarrow} c_{k'\uparrow}) (u_k + v_k e^{-i\phi_j} c_{-k\downarrow} c_{k\uparrow}) c_{k\uparrow}^\dagger c_{k\uparrow} (u_k + v_k e^{i\phi_{j'}} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) \\ &\quad \times \prod_{k'' \neq k} (u_{k''} + v_{k''} e^{i\phi_{j'}} c_{k''\uparrow}^\dagger c_{-k''\downarrow}^\dagger) |0\rangle \\ &= \langle 0 | \prod_{k' \neq k} (u_{k'} + v_{k'} e^{-i\phi_j} c_{-k'\downarrow} c_{k'\uparrow}) \cdot v_k^2 e^{i\varphi} \cdot \prod_{k'' \neq k} (u_{k''} + v_{k''} e^{i\phi_{j'}} c_{k''\uparrow}^\dagger c_{-k''\downarrow}^\dagger) |0\rangle \\ &= e^{i\varphi} v_k^2 \frac{\langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle}{\langle 0 | (u_k + v_k e^{-i\phi_j} c_{-k\downarrow} c_{k\uparrow}) (u_k + v_k e^{i\phi_{j'}} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) |0\rangle} \\ &= \frac{v_k^2}{u_k^2 + v_k^2 e^{i\varphi}} e^{i\varphi} \mathcal{W}(\varphi) \end{aligned} \quad (\text{B1})$$

and

$$\begin{aligned}
& \langle \Psi(\phi_j) | c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger c_{-k'\downarrow} c_{k'\uparrow} | \Psi(\phi_{j'}) \rangle \\
&= \langle 0 | \prod_{\kappa \neq k, k'} (u_\kappa + v_\kappa e^{-i\phi_j} c_{-\kappa\downarrow} c_{\kappa\uparrow}) (u_k + v_k e^{-i\phi_j} c_{-k\downarrow} c_{k\uparrow}) c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger (u_k + v_k e^{i\phi_{j'}} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) \\
&\quad \times (u_{k'} + v_{k'} e^{-i\phi_j} c_{-k'\downarrow} c_{k'\uparrow}) c_{-k'\downarrow} c_{k'\uparrow} (u_{k'} + v_{k'} e^{i\phi_{j'}} c_{k'\uparrow}^\dagger c_{-k'\downarrow}^\dagger) \prod_{\kappa'' \neq k, k'} (u_{\kappa''} + v_{\kappa''} e^{i\phi_{j'}} c_{\kappa''\uparrow}^\dagger c_{-\kappa''\downarrow}^\dagger) | 0 \rangle \\
&= u_k v_k u_{k'} v_{k'} e^{i\varphi} \frac{\langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle}{\langle 0 | (u_k + v_k e^{-i\phi_j} c_{-k\downarrow} c_{k\uparrow}) (u_k + v_k e^{i\phi_{j'}} c_{k\uparrow}^\dagger c_{-k\downarrow}^\dagger) (u_{k'} + v_{k'} e^{-i\phi_j} c_{-k'\downarrow} c_{k'\uparrow}) (u_{k'} + v_{k'} e^{i\phi_{j'}} c_{k'\uparrow}^\dagger c_{-k'\downarrow}^\dagger) | 0 \rangle} \\
&= \left(\frac{u_k v_k}{u_k^2 + v_k^2 e^{i\varphi}} \right)^2 e^{i\varphi} \mathcal{W}(\varphi),
\end{aligned} \tag{B2}$$

where $\varphi = \phi_{j'} - \phi_j$, we compute the matrix elements of PH_AP as follows:

$$\begin{aligned}
\langle \Psi(\phi_j) | H_A | \Psi(\phi_{j'}) \rangle &= 2 \sum_k \frac{(\epsilon_k - \mu_A) v_k^2}{u_k^2 + v_k^2 e^{i\varphi}} e^{i\varphi} \mathcal{W}(\varphi) - |g|^2 \left(\sum_k \frac{u_k v_k}{u_k^2 + v_k^2 e^{i\varphi}} \right)^2 e^{i\varphi} \mathcal{W}(\varphi) \\
&= 2e^{i\varphi} \mathcal{W}(\varphi) \sum_k \frac{\sqrt{E_k^2 - \Delta^2} (E_k - \sqrt{E_k^2 - \Delta^2})}{E_k (1 + e^{i\varphi}) + \sqrt{E_k^2 - \Delta^2} (1 - e^{i\varphi})} \\
&\quad - |g|^2 e^{i\varphi} \mathcal{W}(\varphi) \left(\Delta \sum_k \frac{1}{E_k (1 + e^{i\varphi}) + \sqrt{E_k^2 - \Delta^2} (1 - e^{i\varphi})} \right)^2 \\
&\approx e^{i\varphi} \mathcal{W}(\varphi) \int_{\Delta}^{b\Delta} dE \rho(E) \frac{4e^{-i\varphi} E (\Delta^2 - E^2)}{2E^2 - \Delta^2 (1 - \cos(\varphi))} \\
&\quad - |g|^2 e^{i\varphi} \mathcal{W}(\varphi) \left(\int_{\Delta}^{b\Delta} dE \rho(E) \frac{\Delta E (1 + e^{-i\varphi})}{2E^2 - \Delta^2 (1 - \cos(\varphi))} \right)^2,
\end{aligned} \tag{B3}$$

$$\equiv \mathcal{K}(\varphi) - |g|^2 \mathcal{V}(\varphi). \tag{B4}$$

The complex functions $\mathcal{K}(\varphi)$ and $\mathcal{V}(\varphi)$ are

$$\mathcal{K}(\varphi) = n \Delta \mathcal{W}(\varphi) (2 \ln(2b) \cos(\varphi) + \varphi \sin(\varphi) - 2b^2) / (4b) + O(b^{-2}), \tag{B5}$$

$$\mathcal{V}(\varphi) = n^2 \mathcal{W}(\varphi) (2 \ln(2b) \cos(\varphi/2) + \varphi \sin(\varphi/2))^2 / (4b)^2 + O(b^{-3}). \tag{B6}$$

In addition, the diagonal elements of $\mathcal{K}$ and $\mathcal{V}$ are

$$\mathcal{K}(0) = n \Delta (\ln(2b) - b^2) / (2b), \tag{B7}$$

$$\mathcal{V}(0) = (n \ln(2b))^2 / (2b)^2, \tag{B8}$$

where we retain only terms at leading orders in $1/b$.

APPENDIX C: EFFECTIVE DIMENSIONALITY

Here, we give a two-dimensional toy example of the effective dimensionality, to provide some intuition. We define normalized basis vectors $a_1 = \{\cos(\theta), \sin(\theta)\}$ and $a_2 = \{\sin(\theta), \cos(\theta)\}$ with $0 < \theta < \pi/2$, illustrated in Fig. 13(a). These vectors are generally linearly independent and nonorthogonal, except at $\theta = \pi/4$ where $a_1 = a_2$. At $\theta = 0$, the basis is orthonormal.

For $\theta = \pi/4 - \varepsilon$ with $0 < \varepsilon \ll \pi/4$, these two vectors are highly overlapped. While they are strictly linearly inde-

pendent and span a two-dimensional space, intuitively, any “reasonable” linear combination (by which we mean linear combinations with coefficients of order 1) of the two vectors will be close to the 1D subspace $x = y$ spanned by the vector $a_1 + a_2 \propto [1, 1]$. To move far from the manifold $x = y$ requires large-amplitude coefficients of the nonorthogonal basis states. These amplitudes are strongly dependent on ε , reflecting the ill-conditioned overlap matrix.

In this simple example, the basis overlap is $a_1 \cdot a_2 = \sin(2\theta) = \cos(2\varepsilon)$, and the overlap matrix is the 2×2 circulant matrix generated by $\{1, \sin(2\theta)\}$, with eigenvalues

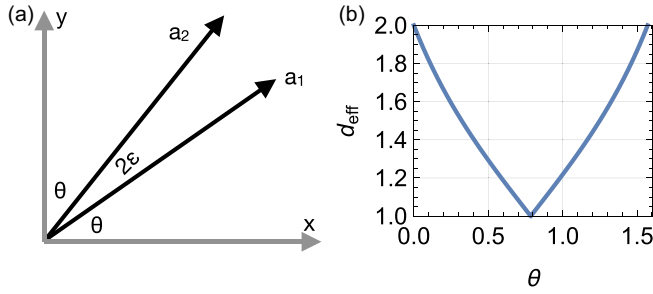


FIG. 13. (a) A cartoon of two nearly colinear basis states a_1 and a_2 . (b) The effective dimensionality computed using Eq. (36), showing a continuous interpolation from $d_{\text{eff}} = 1$ to $d_{\text{eff}} = 2$ as the basis changes from colinear at $\theta = \pi/4$ to orthonormal at $\theta = 0, \pi/2$.

$\lambda_{\pm} = 1 \pm \sin(2\theta)$. We use the definition of the effective dimensionality in Eq. (36) to compute

$$d_{\text{eff}} = 1 + \sqrt{\lambda_-/\lambda_+}. \quad (\text{C1})$$

This functional form is shown in Fig. 13(b). It is straightforward to verify the limits: $d_{\text{eff}} = 1$ when $\theta = \pi/4$ (i.e., when $a_1 = a_2$ are colinear), and $d_{\text{eff}} = 2$ when $\theta = 0$ or $\pi/2$ (i.e., when $a_{1,2}$ are orthogonal), conforming to the usual integer-valued dimensionalities in these cases. This 2D example is illustrative of the reduced effective dimensionality of nearly colinear basis states.

APPENDIX D: JOSEPHSON EFFECT AND GREEN'S FUNCTIONS

We use Green's functions to calculate the second-order perturbation term. For a given diagonalizable Hamiltonian, the Fourier transform of its propagator relates to its Hamiltonian inverse. That is,

$$H_T \frac{1}{E_0 \bar{P} - \bar{P} H_0 \bar{P}} H_T D_\varphi = \mathcal{F}_{\tau \rightarrow E_0} [H_T U_0(\tau) H_T D_\varphi],$$

where $U_0(\tau)$ is the propagator for H_0 at time τ . The matrix elements of the projection in Eq. (65) are then the Fourier transformation of the expectation value:

$$\begin{aligned} \langle \Psi(\phi_j) | H_T (E_0 \bar{P} - \bar{P} H_0 \bar{P})^{-1} H_T D_\varphi | \Psi(\phi_j) \rangle \\ = \mathcal{F}_{\tau \rightarrow E_0} \langle H_T U_0(\tau) H_T D_\varphi \rangle. \end{aligned} \quad (\text{D1})$$

The expectation value on the right-hand side is taken with respect to the BCS ground state:

$$\begin{aligned} \langle H_T(\tau) H_T D_\varphi \rangle &= \langle \Psi(\phi_j) | H_T(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle \\ &= \langle \Psi(\phi_j) | U_0^\dagger(\tau) H_T U_0(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle \\ &= e^{iE_0 \tau} \langle \Psi(\phi_j) | H_T U_0(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle, \end{aligned}$$

so

$$\langle H_T U_0(\tau) H_T D_\varphi \rangle = e^{-iE_0 \tau} \langle \Psi(\phi_j) | H_T(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle. \quad (\text{D2})$$

Taking the Fourier transform on the both sides,

$$\begin{aligned} \mathcal{F}_{\tau \rightarrow E_0} [\langle \Psi(\phi_j) | H_T U_0(\tau) H_T | \Psi(\phi_j) \rangle] &= \mathcal{F}_{\tau \rightarrow E_0} [e^{-iE_0 \tau} \langle \Psi(\phi_j) | H_T(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle] \\ &= \mathcal{F}_{\tau \rightarrow E_0} [\langle \Psi(\phi_j) | U_0(\tau) H_T(\tau) H_T D_\varphi | \Psi(\phi_j) \rangle] \\ &= \mathcal{F}_{\tau \rightarrow E_0} [\langle \Psi(\phi_j) | \mathcal{T}_\tau (H_T(\tau) H_T D_\varphi) | \Psi(\phi_j) \rangle] \\ &= \mathcal{F}_{\tau \rightarrow E_0} [\mathcal{T}_\tau \langle H_T(\tau) H_T D_\varphi \rangle], \end{aligned} \quad (\text{D3})$$

i.e., the matrix elements are the Fourier transform of the Green's functions $\mathcal{F}_{\tau \rightarrow E_0} \langle \mathcal{T}_\tau (H_T(\tau) H_T D_\varphi) \rangle$.

The propagator $U_0(\tau)$ can be expanded as the time-ordered series:

$$U_0(\tau) = \mathcal{T}_\tau \exp \left(-\frac{i}{\hbar} \int_0^\tau dt H_0(t) \right) = \mathcal{T}_\tau \left(\mathbb{I} - \frac{i}{\hbar} \int_0^\tau dt H_0(t) + \dots \right),$$

such that

$$U_0(\tau) H_T(\tau) H_T D_\varphi = \mathcal{T}_\tau \left(H_T(\tau) H_T D_\varphi - \frac{i}{\hbar} \int_0^\tau dt H_0(t) H_T(\tau) H_T D_\varphi + \dots \right) \approx \mathcal{T}_\tau (H_T(\tau) H_T D_\varphi).$$

Expanding the time-ordered product in Eq. (D3) and applying Wick's theorem to factorize the correlation functions, we have

$$\begin{aligned} \langle \mathcal{T}_\tau c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_B s}(\tau) c_{\mathbf{k}_A \sigma}^\dagger c_{\mathbf{k}_B \sigma} D_{\varphi_A} D_{\varphi_B} \rangle &= \theta(\tau) \langle c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_B s}(\tau) c_{\mathbf{k}_A \sigma}^\dagger c_{\mathbf{k}_B \sigma} D_{\varphi_A} D_{\varphi_B} \rangle - \theta(-\tau) \langle c_{\mathbf{k}_A \sigma}^\dagger c_{\mathbf{k}_B \sigma} D_{\varphi_A} D_{\varphi_B} c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_B s}(\tau) \rangle \\ &= 2\theta(\tau) \langle c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_A \sigma}^\dagger D_{\varphi_A} \rangle \langle c_{\mathbf{k}_B s}(\tau) c_{\mathbf{k}_B \sigma} D_{\varphi_B} \rangle - 2\theta(-\tau) \langle c_{\mathbf{k}_A \sigma}^\dagger D_{\varphi_A} c_{\mathbf{k}_A s}^\dagger(\tau) \rangle \langle c_{\mathbf{k}_B \sigma} D_{\varphi_B} c_{\mathbf{k}_B s}(\tau) \rangle \\ &= 2 \langle \mathcal{T}_\tau c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_A \sigma}^\dagger D_{\varphi_A} \rangle \langle \mathcal{T}_\tau c_{\mathbf{k}_B \sigma} D_{\varphi_B} c_{\mathbf{k}_B s}(\tau) \rangle \\ &= 2 \langle \mathcal{T}_\tau c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_A \sigma}^\dagger D_{\varphi_A} \rangle \langle \mathcal{T}_\tau c_{\mathbf{k}_B \sigma}(-\tau) c_{\mathbf{k}_B s} D_{\varphi_B} \rangle, \end{aligned} \quad (\text{D4})$$

where subscripts s, σ denote the spin indices, and the expectation values are taken with respect to $|\Psi(\phi_j)\rangle$. With the factorization of the expectation value (proved in Appendix D 1)

$$\langle c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_A \sigma}^\dagger (N_A^e)^n \rangle = \langle c_{\mathbf{k}_A s}^\dagger(\tau) c_{\mathbf{k}_A \sigma}^\dagger \rangle \langle (N_A^e)^n \rangle, \quad (\text{D5})$$

and the BCS overlap expressed with the phase displacement operator

$$\sum_n \frac{(i\varphi)^n}{n!} \langle (N^e)^n \rangle = \mathcal{W}(\varphi) \quad \varphi = \phi' - \phi, \quad (\text{D6})$$

we expand the phase displacement operator in the first term in Eq. (D4) as a Taylor's series:

$$\begin{aligned} \langle \mathcal{T}_\tau c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger D_{\varphi_A} \rangle &= \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger \rangle + (i\varphi_A/2) \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger (2N_A^e) \rangle + \frac{(i\varphi_A/2)^2}{2!} \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger (2N_A^e)^2 \rangle + \dots \\ &= \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger \rangle + (i\varphi_A) \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger N_A^e \rangle + \frac{(i\varphi_A)^2}{2!} \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger (N_A^e)^2 \rangle + \dots, \end{aligned} \quad (\text{D7a})$$

$$= \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger N_A^e \rangle \cdot \left(1 + (i\varphi_A) \langle N_A^e \rangle + \frac{(i\varphi_A)^2}{2!} \langle (N_A^e)^2 \rangle + \dots \right), \quad (\text{D7b})$$

$$= \langle c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger \rangle \mathcal{W}(\varphi_A), \quad (\text{D7c})$$

where $\mathcal{W}(\varphi_A) = \langle \Psi_A(\phi_j) | \Psi_A(\phi_{j'}) \rangle$ is the overlap function of the BCS ground state for island A. Similarly, for island B,

$$\langle \mathcal{T}_\tau c_{\mathbf{k}_{B\sigma}}(-\tau) c_{\mathbf{k}_{B\sigma}} D_{\varphi_B} \rangle = \langle c_{\mathbf{k}_{B\sigma}}(-\tau) c_{\mathbf{k}_{B\sigma}} \rangle \mathcal{W}(\varphi_B).$$

We have computed the BCS overlap function $\mathcal{W}(\varphi)$ directly from the definition in Sec. II C. Here, with the introduction of the phase displacement operator D_φ , we will do the inner product again to obtain additional insight. The overlap now becomes the expectation value of D_φ :

$$\begin{aligned} \mathcal{W}(\varphi) &= \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle \\ &= \langle \Psi(\phi_j) | D_\varphi | \Psi(\phi_{j'}) \rangle \\ &= \langle 0 | \prod_{q'} (u_{q'} + v_{q'} e^{-i\phi_j} c_{-q'\downarrow} c_{q'\uparrow}) \left[\mathbb{I} + (i\varphi/2) \left(\sum_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{\mathbf{k}\uparrow} + c_{-\mathbf{k}\downarrow}^\dagger c_{-\mathbf{k}\downarrow} \right) + \dots \right] \prod_{q'} (u_{q'} + v_{q'} e^{i\phi_{j'}} c_{q'\uparrow}^\dagger c_{-q'\downarrow}^\dagger) | 0 \rangle, \\ &= 1 + (i\varphi/2) \langle N \rangle + \frac{(i\varphi/2)^2}{2!} \langle N^2 \rangle + \dots, \end{aligned} \quad (\text{D8a})$$

$$= 1 + (i\varphi) \langle N^e \rangle + \frac{(i\varphi)^2}{2!} \langle (N^e)^2 \rangle + \dots, \quad (\text{D8b})$$

$$= \sum_{n=0}^{\infty} \frac{(i\varphi)^n}{n!} \langle (N^e)^n \rangle, \quad (\text{D8c})$$

$$= \langle e^{i\varphi \hat{N}^{\text{el}}} \rangle, \quad (\text{D8d})$$

i.e., we can express $\mathcal{W}(\varphi)$ as the expectation value of D_φ .

Taking the tunneling matrix element as a constant $t_{\mathbf{k}_A \mathbf{k}_B} \approx t$, we obtain

$$\begin{aligned} \langle \mathcal{T}_\tau H_T(\tau) H_T D_\varphi \rangle &= 2 \sum_{\mathbf{k}_A \mathbf{k}_B s \sigma} e^{i\phi_j} t_{\mathbf{k}_A \mathbf{k}_B}^2 \langle \mathcal{T}_\tau c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger \rangle \langle \mathcal{T}_\tau c_{\mathbf{k}_{B\sigma}}(-\tau) c_{\mathbf{k}_{B\sigma}} \rangle \mathcal{W}(\varphi_A) \mathcal{W}(\varphi_B) + \text{c.c.} \\ &= 2 \sum_{\mathbf{k}_A \mathbf{k}_B s \sigma} e^{i\phi_j} t_{\mathbf{k}_A \mathbf{k}_B}^2 \langle \mathcal{T}_\tau c_{\mathbf{k}_{A\sigma}}^\dagger(\tau) c_{\mathbf{k}_{A\sigma}}^\dagger \rangle \langle \mathcal{T}_\tau c_{\mathbf{k}_{B\sigma}}(-\tau) c_{\mathbf{k}_{B\sigma}} \rangle \mathcal{W}(\varphi) + \text{c.c.} \\ &= 2e^{i\phi_j} \sum_{\mathbf{k}_A \mathbf{k}_B s \sigma} t_{\mathbf{k}_A \mathbf{k}_B}^2 \mathcal{F}_{s\sigma}(\mathbf{k}_A, \tau) \mathcal{F}_{s\sigma}^*(\mathbf{k}_B, -\tau) \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle + \text{c.c.} \\ &= 2e^{i\phi_j} \sum_{\mathbf{k}_A \mathbf{k}_B} t_{\mathbf{k}_A \mathbf{k}_B}^2 \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, \tau) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, -\tau) \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle + \text{c.c.} \\ &\approx 2e^{i\phi_j} t^2 \sum_{\mathbf{k}_A \mathbf{k}_B} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, \tau) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, -\tau) \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle + \text{c.c.}, \end{aligned} \quad (\text{D9})$$

which is provided in Eq. (65).

Note that the anomalous Green's functions $\mathcal{F}_{\uparrow\uparrow} = \mathcal{F}_{\downarrow\downarrow} = 0$, as we now show. For simplicity, we consider only a single mode existing on the given superconductor. By Bogoliubov transformation,

$$\begin{pmatrix} b_{\mathbf{k}\uparrow}^\dagger \\ b_{-\mathbf{k}\downarrow} \end{pmatrix} = \begin{pmatrix} u_{\mathbf{k}} & -v_{\mathbf{k}} e^{-i\phi} \\ v_{\mathbf{k}} e^{i\phi} & u_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger \\ c_{-\mathbf{k}\downarrow} \end{pmatrix}. \quad (\text{D10})$$

The time-dependent fermionic annihilation operator is given by

$$\begin{aligned} c_{k_l\uparrow}(\tau) &= e^{H_l\tau} (u_{k_l} b_{k_l\uparrow} + v_{k_l} e^{i\phi} b_{-k_l\downarrow}^\dagger) e^{-H_l\tau} \\ &= u_{k_l} b_{k_l\uparrow} e^{-E_{k_l}\tau} + v_{k_l} e^{i\phi} b_{-k_l\downarrow}^\dagger e^{E_{k_l}\tau} \end{aligned} \quad (D11a)$$

$$\begin{aligned} &= u_{k_l} (u_{k_l} c_{k_l\uparrow} - v_{k_l} e^{i\phi} c_{-k_l\downarrow}^\dagger) e^{-E_{k_l}\tau} + v_{k_l} e^{i\phi} (v_{k_l} e^{-i\phi} c_{k_l\uparrow} + u_{k_l} c_{-k_l\downarrow}^\dagger) e^{E_{k_l}\tau} \\ &= (u_{k_l}^2 e^{-E_{k_l}\tau} + v_{k_l}^2 e^{E_{k_l}\tau}) c_{k_l\uparrow} + u_{k_l} v_{k_l} e^{i\phi} (e^{E_{k_l}\tau} - e^{-E_{k_l}\tau}) c_{-k_l\downarrow}^\dagger. \end{aligned} \quad (D11b)$$

H_l is the superconductor Hamiltonian for the island l . The Green's functions are then

$$\begin{aligned} \langle c_{k_l\uparrow}(\tau) c_{k_l\uparrow} \rangle &= (u_{k_l}^2 e^{-E_{k_l}\tau} + v_{k_l}^2 e^{E_{k_l}\tau}) \langle c_{k_l\uparrow} c_{k_l\uparrow} \rangle + u_{k_l} v_{k_l} e^{i\phi} (e^{E_{k_l}\tau} - e^{-E_{k_l}\tau}) \langle c_{k_l\uparrow} c_{-k_l\downarrow}^\dagger \rangle \\ &= 0, \end{aligned} \quad (D12)$$

$$\langle c_{k_l\uparrow} c_{k_l\uparrow}(\tau) \rangle = 0. \quad (D13)$$

The result can be generalized to the spin-down case:

$$\langle c_{-k_l\downarrow}(\tau) c_{-k_l\downarrow} \rangle = \langle c_{-k_l\downarrow} c_{-k_l\downarrow}(\tau) \rangle = 0.$$

The electron mode k can be arbitrary. Therefore,

$$\mathcal{F}_{ss} \equiv \theta(\tau) \langle c_{ks}(\tau) c_{ks} \rangle - \theta(-\tau) \langle c_{ks} c_{ks}(\tau) \rangle = 0. \quad (D14)$$

The anomalous Green's functions with the identical spin indices vanish: $\mathcal{F}_{\uparrow\uparrow} = \mathcal{F}_{\downarrow\downarrow} = 0$.

1. Factorization in Green's functions

We prove Eq. (D5) here. The expectation value $\langle (N^e)^n \rangle$ is given by

$$\begin{aligned} \langle (N^e)^n \rangle &= \langle 0 | \prod_{q'} (u_{q'} + v_{q'} e^{-i\phi_j} c_{-q'\downarrow} c_{q'\uparrow}) \left(\sum_k c_{k\uparrow}^\dagger c_{k\uparrow} + c_{-k\downarrow}^\dagger c_{-k\downarrow} \right)^n \prod_{q'} (u_{q'} + v_{q'} e^{i\phi_{j'}} c_{q'\uparrow}^\dagger c_{-q'\downarrow}^\dagger) | 0 \rangle \\ &= 2^n \sum_k \langle 0 | \prod_{qq'} (u_{q'} + v_{q'} e^{-i\phi_j} c_{-q'\downarrow} c_{q'\uparrow}) \\ &\quad \times \left\{ (c_{k\uparrow}^\dagger c_{k\uparrow})^n + \left(\sum_{k_1 \neq k} c_{k_1\uparrow}^\dagger c_{k_1\uparrow} c_{k\uparrow}^\dagger c_{k\uparrow} \cdots c_{k\uparrow}^\dagger c_{k\uparrow} \right) + \left(\sum_{k_2 \neq k} c_{k_2\uparrow}^\dagger c_{k_2\uparrow} c_{k\uparrow}^\dagger c_{k\uparrow} \cdots c_{k\uparrow}^\dagger c_{k\uparrow} \right) + \cdots \right\} \\ &\quad \times (u_q + v_q e^{i\phi_{j'}} c_{q\uparrow}^\dagger c_{-q\downarrow}^\dagger) | 0 \rangle \\ &= 2^n \sum_k \left(v_k^{2n} + \sum_{k_1 \neq k} v_{k_1}^2 \underbrace{v_k^2 \cdot s v_k^2}_{n-1} + \sum_{k_2 \neq k} v_{k_2}^2 v_{k_2}^2 \underbrace{s v_k^2}_{n-2} + \cdots \right). \end{aligned} \quad (D15)$$

On the other hand, the correlation functions are

$$\langle c_{k\uparrow}^\dagger(\tau) c_{k\uparrow}^\dagger \rangle = \mathcal{F}_{\uparrow\uparrow}^* = 0, \quad (D16a)$$

$$\langle c_{-k\downarrow}^\dagger(\tau) c_{-k\downarrow}^\dagger \rangle = \mathcal{F}_{\downarrow\downarrow}^* = 0, \quad (D16b)$$

$$\langle c_{k\uparrow}^\dagger(\tau) c_{-k\downarrow}^\dagger \rangle = \mathcal{F}_{\uparrow\downarrow} = u_k v_k^3 (e^{E_k\tau} - e^{-E_k\tau}), \quad (D16c)$$

$$\langle c_{-k\downarrow}^\dagger(\tau) c_{k\uparrow}^\dagger \rangle = \mathcal{F}_{\downarrow\uparrow} = u_k v_k^3 (e^{E_k\tau} - e^{-E_k\tau}). \quad (D16d)$$

The left-hand side of Eq. (D5) is discussed separately. For $s \neq \sigma$, we have

$$\begin{aligned}
 \langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger (N^e)^n \rangle &= 2^n \sum_k \langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger (c_{k\uparrow}^\dagger c_{k\uparrow})^n \rangle + \left\langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger \left(\sum_{k_1 \neq k} c_{k_1\uparrow}^\dagger c_{k_1\uparrow} c_{k\uparrow}^\dagger c_{k\uparrow} \cdots c_{k\uparrow}^\dagger c_{k\uparrow} \right) \right\rangle + \cdots \\
 &= 2^n \sum_k (u_k^2 e^{-E_k \tau} + v_k^2 e^{E_k \tau}) \langle c_{ks}^\dagger c_{k\sigma}^\dagger (e_{k\uparrow}^\dagger c_{k\uparrow})^n \rangle \\
 &\quad + u_k v_k^3 (e^{E_k \tau} - e^{-E_k \tau}) \langle c_{ks}^\dagger c_{k\sigma}^\dagger (c_{k\uparrow}^\dagger c_{k\uparrow})^n \rangle \\
 &\quad + (u_k^2 e^{-E_k \tau} + v_k^2 e^{E_k \tau}) \left\langle c_{ks}^\dagger c_{k\sigma}^\dagger \sum_{k_1 \neq k} c_{k_1\uparrow}^\dagger c_{k_1\uparrow} c_{k\uparrow}^\dagger c_{k\uparrow} \cdots c_{k\uparrow}^\dagger c_{k\uparrow} \right\rangle \\
 &\quad + u_k v_k^3 (e^{E_k \tau} - e^{-E_k \tau}) \left\langle c_{ks}^\dagger c_{k\sigma}^\dagger \sum_{k_1 \neq k} c_{k_1\uparrow}^\dagger c_{k_1\uparrow} c_{k\uparrow}^\dagger c_{k\uparrow} \cdots c_{k\uparrow}^\dagger c_{k\uparrow} \right\rangle \\
 &\quad + \cdots \\
 &= 2^n u_{k'} v_{k'}^3 (e^{E_{k'} \tau} - e^{-E_{k'} \tau}) \sum_k \left(v_k^{2n} + \sum_{k_1 \neq k} v_{k_1}^2 v_k^2 \cdot s v_k^2 + \cdots \right). \tag{D17}
 \end{aligned}$$

However, if $s = \sigma$, all the expectation values vanish according to the exclusion principle,

$$\langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger (N^e)^n \rangle = 0. \tag{D18}$$

Combining the results in Eqs. (D15)–(D18), we obtain

$$\begin{aligned}
 \langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger ((N^e)^n) \rangle &= \langle c_{ks}^\dagger(\tau) c_{k\sigma}^\dagger (N^e)^n \rangle \\
 &= \begin{cases} 0, & s = \sigma, \\ 2^n u_{k'} v_{k'}^3 (e^{E_{k'} \tau} - e^{-E_{k'} \tau}) \sum_k (v_k^{2n} + \sum_{k_1 \neq k} v_{k_1}^2 v_k^2 \cdot v_k^2 + \sum_{k_2 \neq k} v_{k_2}^2 v_k^2 \cdot v_k^2 + \cdots) & s \neq \sigma. \end{cases} \tag{D19}
 \end{aligned}$$

2. Finite-temperature fourier transform

In this section, we compute the matrix elements $\langle \Psi(\phi_j) | H_T (E_0 \bar{P} - \bar{P} H_0 \bar{P})^{-1} H_T | \Psi(\phi_{j'}) \rangle$. First, we take the finite-temperature Fourier transform of Eq. (65) [38], to get

$$\begin{aligned}
 \mathcal{F}_{\tau \rightarrow E_0} \langle H_T(\tau) H_T D_\phi \rangle &= 2e^{i\phi_j} t^2 \frac{1}{\beta} \sum_{i\mathcal{E}_n} \sum_{\mathbf{k}_A \mathbf{k}_B} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, i\mathcal{E}_n) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, i\mathcal{E}_{n'}) \underbrace{\frac{1}{\beta} \int_0^\beta d\tau e^{i\tau(\mathcal{E}_{n'} - \mathcal{E}_n + E_0)} \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle}_{=\delta_{\mathcal{E}_{n'}, \mathcal{E}_n - E_0}} \\
 &= 2e^{i\phi_j} t^2 \sum_{\mathbf{k}_A \mathbf{k}_B} \frac{1}{\beta} \sum_{i\mathcal{E}_n} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, i\mathcal{E}_n) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, i(\mathcal{E}_n - E_0)) \langle \Psi(\phi_j) | \Psi(\phi_{j'}) \rangle + \text{c.c.}, \tag{D20}
 \end{aligned}$$

where β is the Boltzmann temperature and E_0 is the eigenenergy of H_0 . We then compute the Matsubara sum by a contour integral [38]:

$$\begin{aligned}
 \frac{1}{\beta} \sum_{i\mathcal{E}_n} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, i\mathcal{E}_n) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, i(\mathcal{E}_n - E_0)) &= \frac{1}{\beta} \sum_{i\mathcal{E}_n} \frac{-\Delta}{(i\mathcal{E}_n)^2 - ((\epsilon_{\mathbf{k}_A} - \mu_A)^2 + |\Delta|^2)} \frac{-\Delta}{(i\mathcal{E}_n - iE_0)^2 - ((\epsilon_{\mathbf{k}_B} - \mu_B)^2 + |\Delta|^2)} \\
 &\approx \frac{\Delta^2}{4E_{\mathbf{k}_A} E_{\mathbf{k}_B}} \left(\frac{1}{iE_0 + E_{\mathbf{k}_A} + E_{\mathbf{k}_B}} - \frac{1}{iE_0 - E_{\mathbf{k}_A} - E_{\mathbf{k}_B}} \right), \quad \text{for } \beta \rightarrow \infty, \tag{D21a}
 \end{aligned}$$

$$= \frac{\Delta^2}{2E_{\mathbf{k}_A} E_{\mathbf{k}_B}} \cdot \frac{E_{\mathbf{k}_A} + E_{\mathbf{k}_B}}{E_0^2 + (E_{\mathbf{k}_A} + E_{\mathbf{k}_B})^2}. \tag{D21b}$$

The residues occur at the poles $i\mathcal{E}_r = \pm E_{\mathbf{k}_A}, iE_0 \pm E_{\mathbf{k}_B}$. The number of excitation is chosen to be zero as we assume zero temperature $\beta \rightarrow \infty$ in our case. We compute the momentum sum by integral approximation $\sum_{\mathbf{k}=-\infty}^\infty \rightarrow 2 \int_{-\infty}^\infty dE_k \rho(E_k)$. After

some algebra given in Appendix D3, we obtain

$$\sum_{\mathbf{k}_A \mathbf{k}_B} \frac{1}{\beta} \sum_{i\mathcal{E}_n} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, i\mathcal{E}_n) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, i(\mathcal{E}_n - E_0)) = \frac{2N_A^e N_B^e}{b^2 - 1} \mathcal{I}(E_0, \Delta). \quad (\text{D22})$$

We evaluate the double integral $\mathcal{I}(E_0, \Delta)$ in Appendix D3.

3. Double Integral $\mathcal{I}(E_0, \Delta)$

Given the Matsubara sum in Eq. (D21b) and applying the integral approximation to the momentum sums, Eq. (D22) becomes

$$\begin{aligned} \sum_{\mathbf{k}_A \mathbf{k}_B} \frac{1}{\beta} \sum_{i\mathcal{E}_n} \mathcal{F}_{\uparrow\downarrow}(\mathbf{k}_A, i\mathcal{E}_n) \mathcal{F}_{\uparrow\downarrow}^*(\mathbf{k}_B, i(\mathcal{E}_n - E_0)) &= \int_{\Delta}^{\infty} dE_{k_A} \int_{\Delta}^{\infty} dE_{k_B} \rho(E_{k_A}) \rho(E_{k_B}) \frac{\Delta^2}{2E_{k_A} E_{k_B} E_0^2 + (E_{k_A} + E_{k_B})^2} \\ &\approx \frac{N_A^e N_B^e}{2(b^2 - 1)} \int_{\Delta}^{\infty} \int_{\Delta}^{\infty} \frac{(E_{k_A} + E_{k_B}) dE_{k_A} dE_{k_B}}{\sqrt{(E_{k_A}^2 - \Delta^2)(E_{k_B}^2 - \Delta^2)(E_0^2 + (E_{k_A} + E_{k_B})^2)}} + \mathcal{O}(E_0/\Delta) \\ &\equiv \frac{N_A^e N_B^e}{2(b^2 - 1)} \mathcal{I}(E_0, \Delta), \end{aligned} \quad (\text{D23})$$

in which we approximate the momentum sums by a double integral $\mathcal{I}(E_0, \Delta)$. Here, we compute $\mathcal{I}(E_0, \Delta)$ in two limits: (1) $E_0 \ll \Delta$ and (2) $E_0 \gg \Delta$.

a. $E_0 \ll \Delta$. We expand the integrand around $E_0 = 0$:

$$\begin{aligned} \mathcal{I} &\approx \int_{\Delta}^{\infty} dE_{k_A} \int_{\Delta}^{\infty} dE_{k_B} \left(\frac{1}{(E_{k_A} + E_{k_B}) \sqrt{(E_{k_A}^2 - \Delta^2)(E_{k_B}^2 - \Delta^2)}} - \frac{E_0^2}{(E_{k_A} + E_{k_B})^3 \sqrt{(E_{k_A}^2 - \Delta^2)(E_{k_B}^2 - \Delta^2)}} \right) + \mathcal{O}\left(\frac{E_0}{\Delta}\right)^3 \\ &= \frac{-\pi(E_0^2 - 16\Delta^2)}{64\Delta^3} + \mathcal{O}\left(\frac{E_0}{\Delta}\right)^4, \quad \text{for } E_0 \ll \Delta. \end{aligned}$$

So,

$$\mathcal{I}(E_0, \Delta) \approx \frac{\pi^2}{4\Delta} \left(1 - \frac{E_0^2}{16\Delta^2} \right), \quad \text{for } E_0 \ll \Delta. \quad (\text{D24})$$

b. $E_0 \gg \Delta$. We expand the integral around the limit $E_0 \rightarrow \infty$. Introducing the triangle transformation in Fig. 14(a) with $E_{k_A} \equiv \Delta/\sin\theta_A$ and $E_{k_B} \equiv \Delta/\sin\theta_B$, the integral changes: $\int_{\Delta}^{\infty} dE_{k_A} \int_{\Delta}^{\infty} dE_{k_B} \rightarrow \int_0^{\pi/2} d\theta_A (-\Delta \cot\theta_A \csc\theta_A) \int_0^{\pi/2} d\theta_B (-\Delta \cot\theta_B \csc\theta_B)$. Rescaling the energy by Δ , we

obtain

$$\begin{aligned} \mathcal{I}(E_0, \Delta) &= \int_0^{\pi/2} \int_0^{\pi/2} d\theta_A d\theta_B \frac{\csc^2\theta_A \csc^2\theta_B}{E_0^2 + \Delta^2(\csc\theta_A + \csc\theta_B)^2} \\ &\approx \frac{1}{\Delta} \cdot \begin{cases} \int d\theta_B \cdot 1 & \text{for } \theta_B \ll 1, \\ \int d\theta_B \frac{\csc\theta_B(\pi E_0 - 2 \csc\theta_B)}{2E_0^2} & \text{for } E_0 \gg \Delta. \end{cases} \end{aligned} \quad (\text{D25})$$

A crossover approximation around $\theta_B = 1/E_0$ is dealt with by integrating the two approximations in Eq. (D25) up to $\mathcal{O}(E_0)$:

$$\begin{aligned} \mathcal{I}(E_0, \Delta) &\approx \frac{1}{\Delta} \left(\int_0^{1/E_0} d\theta_B + \int_{1/E_0}^{\pi/2} d\theta_B \frac{\csc\theta_B(\pi E_0 - 2 \csc\theta_B)}{2E_0^2} \right) \\ &= \frac{\pi \ln(2E_0/\Delta)}{2E_0} + \mathcal{O}\left(\frac{E_0}{\Delta}\right)^3, \quad \text{for } E_0 \gg \Delta. \end{aligned} \quad (\text{D26})$$

Combining Eqs. (D24) and (D26), we obtain the double integral approximation for the double sum in Eq. (D22):

$$\mathcal{I}(E_0, \Delta) = \begin{cases} \frac{\pi^2}{2\Delta} \left(1 - \frac{E_0^2}{16\Delta^2} \right), & \text{for } E_0 \ll \Delta, \\ \frac{\pi \ln(2E_0/\Delta)}{2E_0}, & \text{for } E_0 \gg \Delta. \end{cases}$$

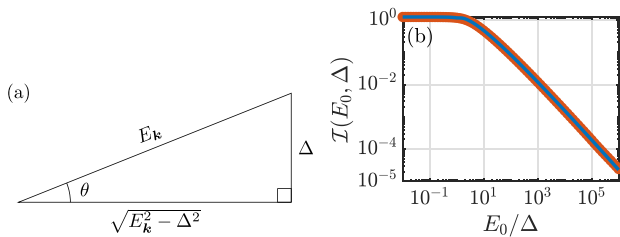


FIG. 14. (a) Triangle transformation for calculating the momentum sum in $\mathcal{I}(E_0, \Delta)$. With this transformation, we change the integral $\int_0^{\infty} dE_k \rightarrow \int_0^{\pi/2} d\theta (-\Delta \cot\theta \csc\theta)$. (b) Numerical (circles) and analytical approximation (solid line) of the double sum $\mathcal{I}(E_0, \Delta)$.

Figure 14(b) shows the analytical solution $\mathcal{I}(E_0, \Delta)$ (solid line) and the numerical double sum (circle markers), indicating good agreement.

APPENDIX E: SPATIALLY VARYING BCS THEORY

The inductance of an inductor depends on its geometry and material properties. Consider a superconductor volume “SC” in the presence of a vector potential $\mathbf{A}$. The Hamiltonian is given by [24,66]

$$H_{K_A} + H_{I_A} = \int_{\text{SC}} dV \left\{ c_{rs}^\dagger \left[\frac{1}{2m} (\mathbf{p} - e\mathbf{A}^{\text{int}})^2 - \mu \right] c_{rs} - |g|^2 c_{r\uparrow}^\dagger c_{-r\downarrow}^\dagger c_{-r'\downarrow} c_{r'\uparrow} \right\}. \quad (\text{E1})$$

In the Hamiltonian, the electron momentum operator $\mathbf{p}$ acts on the fermionic Hilbert space, while the vector potential operator $\mathbf{A}^{\text{int}}$ acts on the electromagnetic Hilbert space. However, in the low-frequency limit, the $\mathbf{A}^{\text{int}}$ field will be determined by the local current density, according to London’s equation. The fermionic field operators in the spatial representation are the Fourier transform of those in the momentum space [24]:

$$c_{rs} = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{ks}. \quad (\text{E2})$$

The matter phase of the superconductor now has position dependence, so the basis state depends on the position as well. The system projector onto the low-energy subspace becomes

$$\Pi = \bigotimes_r P_r, \quad (\text{E3})$$

where $P_r = \sum_{\phi(r)} |\Psi_r[\phi(r)]\rangle \langle \Psi_r[\phi(r)]|$ is the low-energy projector on the coordinate $\mathbf{r}$. For simplicity, we consider the quasi-1D wire model, so the projector reduces to Eq. (75), depending only on the axial coordinate z . The z -dependent basis state can be expressed with the use of Eq. (E2) and Lemma A 1:

$$|\Psi_z(\phi[z])\rangle = \prod_{\mathbf{k}} [u_{\mathbf{k}} + v_{\mathbf{k}} (e^{i\phi(z)/2} c_{\mathbf{k}\uparrow}^\dagger) (e^{i\phi(z)/2} c_{-\mathbf{k}\downarrow}^\dagger)] |0\rangle. \quad (\text{E4})$$

The derivation is given in Appendix E 1. The inclusion of the spatial dependence gives rise to an additional phase in fermionic field operators. After transforming the fermionic Hamiltonian in Eq. (1) to the spatial representation in Eq. (E1) and constructing the new distributed projector Π in Eq. (75), we can evaluate the inductance Hamiltonian by the projection

$$H_L = \Pi (H_{K_A} + H_{I_A}) \Pi \\ = \bigotimes_z P_z H P_z$$

$$= \bigotimes_z \sum_{\phi(z)\bar{\phi}(z)} \langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\bar{\phi}(z)] \rangle \\ \times |\Psi_z[\phi(z)]\rangle \langle \Psi_z[\bar{\phi}(z)]| \\ \approx \bigotimes_z \sum_{\phi(z)} \langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\phi(z)] \rangle \\ \times |\Psi_z[\phi(z)]\rangle \langle \Psi_z[\phi(z)]|. \quad (\text{E5})$$

In the last line, we apply the approximation

$$\langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\bar{\phi}(z)] \rangle \\ \approx \langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\phi(z)] \rangle \delta_{\phi(z), \bar{\phi}(z)},$$

according to orthogonality of BCS basis states at each point along z . Namely, the off-diagonal matrix elements are negligible in the infinite limit $n \rightarrow \infty$.

For simplicity, we choose the gauge

$$\mathcal{G}_{\phi(z)} = e^{i\phi(z)\hat{N}/2}, \quad (\text{E6})$$

such that $c_{zs}^\dagger \mapsto \tilde{c}_{zs}^\dagger = e^{-i\phi(z)/2} c_{zs}^\dagger$ and $c_{zs} \mapsto \tilde{c}_{zs} = e^{i\phi(z)/2} c_{zs}$. This gauge removes the $\phi(z)$ dependence on the bra and ket, i.e.,

$$\langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\phi(z)] \rangle \\ = \langle \Psi_z | e^{-i\phi(z)/2} (H_{K_A} + H_{I_A}) e^{i\phi(z)/2} | \Psi_z \rangle. \quad (\text{E7})$$

In our quasi-1D model, the electron momentum operator is given by $\mathbf{p} \mapsto -i\hbar\partial_z \mathbf{e}_z$ and the vector potential $\mathbf{A}^{\text{int}} = A_z^{\text{int}} \mathbf{e}_z$. The derivative operators acting on the gauge transformed state are

$$\partial_z e^{i\phi(z)/2} c_{zs} | \Psi_z \rangle = e^{i\phi(z)/2} \frac{i}{2} \phi'(z) c_{zs} | \Psi_z \rangle \quad (\text{E8})$$

and

$$\partial_z^2 e^{i\phi(z)/2} c_{zs} | \Psi_z \rangle = e^{i\phi(z)/2} \left[\left(\frac{i}{2} \phi'(z) \right)^2 + \frac{i}{2} \phi''(z) \right] c_{zs} | \Psi_z \rangle. \quad (\text{E9})$$

Assuming the matter phase is adiabatic, $\phi''(z) = 0$, and choosing Coulomb gauge $\nabla \cdot \mathbf{A}^{\text{int}} = 0$, Eq. (E7) reduces to

$$\langle \Psi_z[\phi(z)] | (H_{K_A} + H_{I_A}) | \Psi_z[\phi(z)] \rangle \\ = \frac{1}{2} \frac{De^2}{m} \int_{\text{SC}} dV \left(\frac{\Phi_0}{2\pi} \phi'(z) - A_z^{\text{int}} \right)^2 + \mathcal{V}(\phi), \quad (\text{E10})$$

where the electron density is defined by

$$D \equiv \langle \Psi_z | c_{zs}^\dagger c_{zs} | \Psi_z \rangle \quad (\text{E11})$$

and $\varphi = \bar{\phi} - \phi$. As the projection of the interaction term gives a constant, we can choose the energy baseline to be $\Pi H_{I_A} \Pi = \mathcal{V}(\varphi) = 0$. The projection then becomes

$$\Pi H_{K_A} \Pi \approx \bigotimes_z \sum_{\phi(z)} \frac{1}{2} \frac{De^2}{m} \int_{\text{SC}} dV \left(\frac{\Phi_0}{2\pi} \phi'(z) - A_z^{\text{int}} \right)^2 |\Psi_z[\phi(z)]\rangle \langle \Psi_z[\phi(z)]| = \frac{1}{2} \frac{1}{\mu_0 \lambda_L^2} \int_{\text{SC}} dV \left(\frac{\Phi_0}{2\pi} \hat{\phi}'(z) - A_z^{\text{int}}(\hat{z}) \right)^2, \quad (\text{E12})$$

in which we define the London penetration depth λ_L via $De^2/m = 1/\mu_0 \lambda_L^2$. Considering a cylindrical superconducting wire, the volume integral over the superconducting device is $\int_{\text{SC}} dV = \int_0^{2\pi} d\theta \int_0^L dz \int_0^{r_w} r dr$, and $A_z^{\text{int}}(\hat{z})$ is given in Eq. (F6). With

the current operator $\hat{i}_s = \frac{\Phi_0}{2\pi} \hat{\phi}'(z)/\bar{L}$ adopted from Eq. (88), and applying the lumped-element approximation, $\hat{\phi}'(z) = (\hat{\phi}_B - \hat{\phi}_A)/l = -\hat{\phi}/l$, Eq. (E12) becomes

$$\begin{aligned} \Pi H_{K_A} \Pi &= \frac{1}{2} \left(\frac{\Phi_0}{2\pi} \right)^2 \frac{1}{\mu_0 \lambda_L^2} 2\pi \underbrace{\int_0^l dz \hat{\phi}^2(z)}_{=\hat{\phi}^2/l} \int_0^{r_w} r dr \left(1 - \frac{\mu_0 \lambda_L}{2\pi r_w \bar{L}} \frac{1 - I_0(r/\lambda_L)}{I_1(r_w/\lambda_L)} \right)^2 \\ &= \frac{1}{2} \left(\frac{\Phi_0}{2\pi} \right)^2 \frac{2\pi}{\mu_0 l} \frac{\hat{\phi}^2}{\lambda_L^2} \left\{ \frac{r_w^2}{2} + \frac{\mu_0 l}{2\pi L} \left[\lambda_L^2 \left(\frac{\mu_0 l}{4\pi L} \left(\frac{1 + I_0^2(r_w/\lambda_L)}{I_1^2(r_w/\lambda_L)} - 1 \right) + 2 \right) - \frac{1}{I_1(r_w/\lambda_L)} \left(r_w \lambda_L - \frac{\mu_0 l}{\pi L} \frac{\lambda_L^3}{r_w} \right) \right] \right\} \\ &\approx \frac{1}{2} \left(\frac{\Phi_0}{2\pi} \right)^2 \hat{\phi}^2 \frac{1}{L}, \quad \text{for } \lambda_L \ll r_w, \end{aligned} \quad (\text{E13})$$

where the $I_1^{-1}(r_w/\lambda_L)$ is exponentially suppressed and $(1 + I_0^2(r_w/\lambda_L))/I_1^2(r_w/\lambda_L) \approx 1$ in the limit $r_w \gg \lambda_L$.

Combining these results, we find

$$H_L = \Pi (H_{K_A} + H_{I_A}) \Pi \approx \frac{E_L}{2} \hat{\phi}^2, \quad (\text{E14})$$

where the inductance energy of the system E_L consists of the kinetic and geometric contributions: $E_L = (\Phi_0/2\pi)^2/L$, where $L = L_K + L_G$.

1. Spatial-dependent BCS state and projections

The BCS state changes when we consider a spatial-varying matter phase $\phi(\mathbf{r})$. Here, we consider the case in which the matter phase depends on all three coordinates of the space. To derive the BCS state in position representation, we require the Fourier's transform in Eq. (E2) and Lemma A 1. With the substitution $u_k \rightarrow A_k$, $v_k e^{i\phi} \int d\mathbf{r}_k \int d\mathbf{r}'_k e^{ik \cdot (\mathbf{r}_k - \mathbf{r}'_k)} c_{\mathbf{r}_k \uparrow}^\dagger c_{-\mathbf{r}'_k \downarrow}^\dagger \rightarrow B_k$, we obtain

$$\begin{aligned} |\Psi(\phi)\rangle &= \prod_k (u_k + v_k e^{i\phi} c_{\mathbf{k} \uparrow}^\dagger c_{-\mathbf{k} \downarrow}^\dagger) |0\rangle \\ &= \prod_k \left(u_k + v_k e^{i\phi} \int d\mathbf{r}_k \int d\mathbf{r}'_k e^{ik \cdot (\mathbf{r}_k - \mathbf{r}'_k)} c_{\mathbf{r}_k \uparrow}^\dagger c_{-\mathbf{r}'_k \downarrow}^\dagger \right) |0\rangle \\ &= \sum_{m=0}^n \sum_{\chi \in S_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} e^{i\phi} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} c_{\mathbf{r}_{\mathbf{k}_j} \uparrow}^\dagger c_{-\mathbf{r}'_{\mathbf{k}_j} \downarrow}^\dagger |0\rangle \\ &= \sum_{m=0}^n \sum_{\chi \in S_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} (e^{i\phi/2} c_{\mathbf{r}_{\mathbf{k}_j} \uparrow}^\dagger) (e^{i\phi/2} c_{-\mathbf{r}'_{\mathbf{k}_j} \downarrow}^\dagger) |0\rangle \\ &\rightarrow \sum_{m=0}^n \sum_{\chi \in S_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} (e^{i\phi(\mathbf{r})/2} c_{\mathbf{r}_{\mathbf{k}_j} \uparrow}^\dagger) (e^{i\phi(\mathbf{r})/2} c_{-\mathbf{r}'_{\mathbf{k}_j} \downarrow}^\dagger) |0\rangle, \end{aligned} \quad (\text{E15a})$$

$$\begin{aligned} &= \prod_k [u_k + v_k (e^{i\phi(\mathbf{r})/2} c_{\mathbf{k} \uparrow}^\dagger) (e^{i\phi(\mathbf{r})/2} c_{-\mathbf{k} \downarrow}^\dagger)] |0\rangle \\ &\equiv |\Psi[\phi(\mathbf{r})]\rangle. \end{aligned} \quad (\text{E15b})$$

In Eq. (E15a), we generalize the matter phase from constant to position-dependent, $\phi \rightarrow \phi(\mathbf{r})$. Hence, we express the BCS ground state in spatial representation.

The additional phase terms in Eq. (E15a) vanish under the gauge transformation $\mathcal{G}_{\phi(\mathbf{r})} = e^{i\phi(\mathbf{r})\hat{N}/2}$. Therefore, the electron kinetic energy with respect to the ground state reads

$$\begin{aligned} &\left\langle \int d\mathbf{r} c_{\mathbf{r}s}^\dagger \left[\frac{1}{2m} (i\hbar \nabla - q\mathbf{A})^2 - \mu \right] c_{\mathbf{r}s} \right\rangle \\ &= \langle \Psi[\phi(\mathbf{r})] | \int d\mathbf{r} c_{\mathbf{r}s}^\dagger \left[\frac{1}{2m} (i\hbar \nabla - q\mathbf{A})^2 - \mu \right] c_{\mathbf{r}s} | \Psi[\phi(\mathbf{r})] \rangle, \end{aligned} \quad (\text{E16a})$$

$$\begin{aligned}
&= \langle 0 | \sum_{m=0}^n \sum_{\chi \in \mathcal{S}_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{-i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} (e^{-i\phi(\mathbf{r})/2} c_{-\mathbf{r}_{\mathbf{k}_j} \downarrow}) (e^{-i\phi(\mathbf{r})/2} c_{\mathbf{r}'_{\mathbf{k}_j} \uparrow}) \\
&\quad \times \int d\mathbf{r} c_{rs}^\dagger \left[\frac{1}{2m} (i\hbar \nabla - q\mathbf{A})^2 - \mu \right] c_{rs} \sum_{m'=0}^n \sum_{\chi' \in \mathcal{S}'_m} \prod_{\mathbf{k}'_i \in \chi'^c} u_{\mathbf{k}'_i} \prod_{\mathbf{k}'_j \in \chi'} v_{\mathbf{k}'_j} \int d\mathbf{r}''_{\mathbf{k}'_j} \int d\mathbf{r}'''_{\mathbf{k}'_j} \\
&\quad \times e^{i\mathbf{k}'_j \cdot (\mathbf{r}''_{\mathbf{k}'_j} - \mathbf{r}'''_{\mathbf{k}'_j})} (e^{i\phi(\mathbf{r})/2} c_{\mathbf{r}''_{\mathbf{k}'_j} \uparrow}^\dagger) (e^{i\phi(\mathbf{r})/2} c_{-\mathbf{r}'''_{\mathbf{k}'_j} \downarrow}^\dagger) |0\rangle \\
&= \langle 0 | \sum_{m=0}^n \sum_{\chi \in \mathcal{S}_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{-i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} \\
&\quad \times \underbrace{\mathcal{G}_{\phi(\mathbf{r})}^\dagger \mathcal{G}_{\phi(\mathbf{r})} (e^{-i\phi(\mathbf{r})/2} c_{-\mathbf{r}_{\mathbf{k}_j} \downarrow}) \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=c_{-\mathbf{r}_{\mathbf{k}_j} \downarrow}} \underbrace{\mathcal{G}_{\phi(\mathbf{r})} (e^{-i\phi(\mathbf{r})/2} c_{\mathbf{r}'_{\mathbf{k}_j} \uparrow}) \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=c_{\mathbf{k}_j \uparrow}^\dagger} \mathcal{G}_{\phi(\mathbf{r})} \\
&\quad \times \int d\mathbf{r} \underbrace{\mathcal{G}_{\phi(\mathbf{r})}^\dagger \mathcal{G}_{\phi(\mathbf{r})} c_{rs}^\dagger \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=e^{i\phi(\mathbf{r})/2} c_{rs}^\dagger} \mathcal{G}_{\phi(\mathbf{r})} \left[\frac{1}{2m} (i\hbar \nabla - q\mathbf{A})^2 - \mu \right] \underbrace{\mathcal{G}_{\phi(\mathbf{r})}^\dagger \mathcal{G}_{\phi(\mathbf{r})} c_{rs} \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=e^{i\phi(\mathbf{r})/2} c_{rs}} \mathcal{G}_{\phi(\mathbf{r})} \\
&\quad \times \sum_{m'=0}^n \sum_{\chi' \in \mathcal{S}'_m} \prod_{\mathbf{k}'_i \in \chi'^c} u_{\mathbf{k}'_i} \prod_{\mathbf{k}'_j \in \chi'} v_{\mathbf{k}'_j} \int d\mathbf{r}''_{\mathbf{k}'_j} \int d\mathbf{r}'''_{\mathbf{k}'_j} e^{i\mathbf{k}'_j \cdot (\mathbf{r}''_{\mathbf{k}'_j} - \mathbf{r}'''_{\mathbf{k}'_j})} \\
&\quad \times \underbrace{\mathcal{G}_{\phi(\mathbf{r})}^\dagger \mathcal{G}_{\phi(\mathbf{r})} (e^{i\phi(\mathbf{r})/2} c_{\mathbf{r}''_{\mathbf{k}'_j} \uparrow}^\dagger) \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=c_{\mathbf{r}''_{\mathbf{k}'_j} \uparrow}^\dagger} \underbrace{\mathcal{G}_{\phi(\mathbf{r})} (e^{i\phi(\mathbf{r})/2} c_{-\mathbf{r}'''_{\mathbf{k}'_j} \downarrow}^\dagger) \mathcal{G}_{\phi(\mathbf{r})}^\dagger}_{=c_{-\mathbf{r}'''_{\mathbf{k}'_j} \downarrow}^\dagger} \mathcal{G}_{\phi(\mathbf{r})} |0\rangle \\
&= \langle 0 | \sum_{m=0}^n \sum_{\chi \in \mathcal{S}_m} \prod_{\mathbf{k}_i \in \chi^c} u_{\mathbf{k}_i} \prod_{\mathbf{k}_j \in \chi} v_{\mathbf{k}_j} \int d\mathbf{r}_{\mathbf{k}_j} \int d\mathbf{r}'_{\mathbf{k}_j} e^{-i\mathbf{k}_j \cdot (\mathbf{r}_{\mathbf{k}_j} - \mathbf{r}'_{\mathbf{k}_j})} \mathcal{G}_{\phi(\mathbf{r})}^\dagger c_{-\mathbf{r}_{\mathbf{k}_j} \downarrow} c_{\mathbf{r}'_{\mathbf{k}_j} \uparrow} \\
&\quad \times \frac{1}{2m} \int d\mathbf{r} e^{-i\phi(\mathbf{r})/2} c_{rs}^\dagger [(i\hbar \nabla - q\mathbf{A})^2 - \mu] e^{i\phi(\mathbf{r})/2} c_{rs} \\
&\quad \times \sum_{m'=0}^n \sum_{\chi' \in \mathcal{S}'_m} \prod_{\mathbf{k}'_i \in \chi'^c} u_{\mathbf{k}'_i} \prod_{\mathbf{k}'_j \in \chi'} v_{\mathbf{k}'_j} \int d\mathbf{r}''_{\mathbf{k}'_j} \int d\mathbf{r}'''_{\mathbf{k}'_j} e^{i\mathbf{k}'_j \cdot (\mathbf{r}''_{\mathbf{k}'_j} - \mathbf{r}'''_{\mathbf{k}'_j})} c_{\mathbf{r}''_{\mathbf{k}'_j} \uparrow}^\dagger c_{-\mathbf{r}'''_{\mathbf{k}'_j} \downarrow}^\dagger \mathcal{G}_{\phi(\mathbf{r})} |0\rangle \\
&= \langle \Psi(\phi = 0) | \frac{1}{2m} \int d\mathbf{r} e^{-i\phi(\mathbf{r})/2} c_{rs}^\dagger [(i\hbar \nabla - q\mathbf{A})^2 - t\mu] e^{i\phi(\mathbf{r})/2} c_{rs} | \Psi(\phi = 0) \rangle \\
&= \langle \Psi | \frac{1}{2m} \int d\mathbf{r} e^{-i\phi(\mathbf{r})/2} c_{rs}^\dagger [(i\hbar \nabla - q\mathbf{A})^2 - \mu] e^{i\phi(\mathbf{r})/2} c_{rs} | \Psi \rangle. \tag{E16b}
\end{aligned}$$

Note that the derivative operators ∇ and ∇^2 act on the position space, and the field operators $c_{rs}^\dagger$ and c_{rs} act on the fermionic Fock state.

APPENDIX F: VECTOR POTENTIAL AND INDUCTANCE OF A SUPERCONDUCTING WIRE

1. Vector potential of a wire

In this Appendix, we solve the London equation for a superconducting wire in cylindrical coordinates and compute the magnetic field inside the wire, depicted in Fig. 7. We then integrate the flux around the wire out to a cutoff distance, and calculate the geometric inductance as the magnetic field is linked to the current. In addition, we also compute the $\mathbf{A}^{\text{int}}$ field inside the superconductor induced by the supercurrent. This self-induced field inside the superconductor contributes to the overall kinetic inductance.

With Ampère's law, the London equation is expressed as $\nabla \times \nabla \times \mathbf{H}^{\text{int}} + \mathbf{H}^{\text{int}}/\lambda_L^2 = 0$, where $\mathbf{B}^{\text{int}} = \mu_0 \mathbf{H}^{\text{int}}$ and λ_L is London penetration depth. Considering a circular superconducting wire with radius r_w , we have $\mathbf{H}^{\text{int}}(r) = H_\theta(r) \mathbf{e}_\theta$ according to the cylindrical symmetry. The fields inside the metal must be finite and the boundary condition guarantees $\mathbf{H}^{\text{int}}(r = r_w) = H_0 \mathbf{e}_\theta$, so the solution reads

$$\mathbf{H}^{\text{int}}(r) = H_0 \frac{I_1(r/\lambda_L)}{I_1(r_w/\lambda_L)} \mathbf{e}_\theta, \tag{F1}$$

where $I_\alpha(z)$ is the modified Bessel function of the first kind and H_0 is the magnetic field at the surface of the cylinder. From Maxwell's equation (with zero electric field) $\nabla \times \mathbf{H}^{\text{int}} = \mathbf{J}$, the supercurrent density is along the axial direction:

$$\mathbf{J}(r) = \frac{H_0}{\lambda_L} \frac{I_0(r/\lambda_L)}{I_1(r_w/\lambda_L)} \mathbf{e}_z = J_z(r) \mathbf{e}_z. \tag{F2}$$

The supercurrent $\mathbf{J}$ in Eq. (F2) is exponentially suppressed at the center, but is not identically zero. The maximal value is at the surface $\mathbf{J}_{\max} = \mathbf{J}(r_w) = H_0/\lambda_L \mathbf{e}_z$. The total current flowing through the wire is the surface integral over a cross section of the wire:

$$i_s = \int_{\sigma} \mathbf{J} \cdot d\sigma = 2\pi \int_0^{r_w} dr r J_z(r) = 2\pi r_w H_0, \quad (\text{F3})$$

such that the external field outside the wire is given by

$$\mathbf{B}^{\text{ext}}(r) = \frac{\mu_0 i_s}{2\pi r} \mathbf{e}_\theta = \frac{\mu_0 r_w H_0}{r} \mathbf{e}_\theta. \quad (\text{F4})$$

The corresponding vector potential outside the wire is

$$\mathbf{A}^{\text{ext}}(r) = \frac{\mu_0 i_s}{2\pi} \ln \frac{\sqrt{l^2 + r^2} + l}{r} \mathbf{e}_z. \quad (\text{F5})$$

Gauge invariant vector potential inside a superconductor: We choose a gauge in which the internal vector potential is parallel to the supercurrent [36] and satisfies $\mathbf{H}^{\text{int}} = \nabla \times \mathbf{A}^{\text{int}}/\mu_0$. Solving for the current density from Maxwell's equation $\mathbf{J} = \nabla \times \nabla \times \mathbf{A}^{\text{int}}/\mu_0$ gives

$$\begin{aligned} \mathbf{A}^{\text{int}}(r) &= H_0 \mu_0 \lambda_L \frac{1 - I_0(r/\lambda_L)}{I_1(r_w/\lambda_L)} \mathbf{e}_z \\ &= i_s \frac{\mu_0 \lambda_L}{2\pi r_w} \frac{1 - I_0(r/\lambda_L)}{I_1(r_w/\lambda_L)} \mathbf{e}_z \\ &\approx -i_s \frac{\mu_0 \lambda_L e^{(r-r_w)/\lambda_L}}{2\pi \sqrt{r_w r}} \mathbf{e}_z \quad \text{for } \lambda_L \ll r_w. \end{aligned} \quad (\text{F6})$$

We obtain the gauge invariant vector potential inside the superconductor:

$$\tilde{\mathbf{A}}^{\text{int}}(r) = \mathbf{A}^{\text{int}}(r) + \frac{\Phi_0}{2\pi} \nabla \hat{\phi}, \quad (\text{F7})$$

where the scalar ϕ field is self-generated by the superconductor.

2. Wire inductance

There are two ways to compute the inductance L of a wire: (1) Evaluate the integrated flux linking a segment of wire carrying a current i_s , including both the flux internal and external to the wire volume, from which we derive the constitutive law $\Phi_{\text{tot}} = Li_s$, and (2) evaluate the integrated field and kinetic energies, from which we compute the total energy $E_{\text{tot}} = Li_s^2/2$. We now undertake both of these derivations for the classical (i.e., mean field) theory of a superconducting wire, to verify that they both give consistent values for L .

a. Inductance from integrated fluxes. The total flux linking a segment of wire carrying a current i_s includes contributions from the interior and exterior of the wire, which we compute here.

Integrated flux outside the wire. The integrated flux linking the gray surface shown in Fig. 7 on the exterior region of the wire $r > r_w$, over an axial length l , is used to compute the geometric inductance per unit length L_G . The integrated exterior flux is given by

$$\Phi_{\text{ext}} = l \int_{r_w}^R dr |\mathbf{B}^{\text{ext}}(r)| = \underbrace{\frac{\mu_0 l \ln(R/r_w)}{2\pi}}_{=L_G} i_s. \quad (\text{F8})$$

Typically, the cutoff distance R will correspond to the radius of curvature of the wire segment, which forms part of a larger loop.

Integrated flux inside the wire. The flux inside the superconductor decays exponentially, but contributes to the kinetic inductance, which we show here gives $\Phi_{\text{int}} = i_s L_K$. The integrated flux linking the gray surface shown in Fig. 7 on the interior region of the wire $r < r_w$, over an axial length l , is used to compute the geometric inductance per unit length L_K . The integrated interior flux is given by

$$\begin{aligned} \Phi_{\text{int}} &= \int_{S_{\text{int}}} \mathbf{B}^{\text{int}} \cdot d\mathbf{S} \\ &= \frac{\mu_0 i_s l}{2\pi r_w} \frac{1}{I_1(r_w/\lambda_L)} \int_0^{r_w} dr I_1(r/\lambda_L) \\ &= \frac{\mu_0 i_s l}{2\pi r_w} \frac{\lambda_L (I_0(r_w/\lambda_L) - 1)}{I_1(r_w/\lambda_L)} \\ &\approx \frac{\mu_0 l \lambda_L}{2\pi r_w} i_s, \quad \text{for } \lambda_L \ll r_w. \end{aligned} \quad (\text{F9})$$

Given the London penetration depth $\lambda_L = \sqrt{m/(\mu_0 D e^2)}$ and the effective cross-section area $\sigma_{\text{eff}} \approx 2\pi r_w \lambda_L$, we find

$$\Phi_{\text{int}} = \underbrace{\frac{m}{D e^2} \frac{l}{\sigma_{\text{eff}}}}_{\equiv L_K} i_s. \quad (\text{F10})$$

Total integrated flux. The total flux passing through a surface bounded by the centerline of the wire, and extending radially out to a distance R , represented by the gray surface in Fig. 7, is then

$$\Phi_{\text{tot}} = \Phi_{\text{int}} + \Phi_{\text{ext}} = (L_K + L_G) i_s, \quad (\text{F11})$$

from which we see that the total inductance is

$$L = L_K + L_G. \quad (\text{F12})$$

b. Inductance from Energies. Here, we compute the inductance from the field and electron kinetic energies.

Field energy outside the wire. The electromagnetic energy stored in the external field $\mathbf{B}^{\text{ext}}$ is given by

$$E_{\text{ext}} = \frac{1}{2\mu_0} \int dV |\mathbf{B}^{\text{ext}}|^2 = \frac{\mu_0 l \ln(R/r_w)}{2\pi} \frac{i_s^2}{2}. \quad (\text{F13})$$

We identify the prefactor as the geometric inductance, consistent with Eq. (F8).

Electron kinetic energy. The electronic kinetic energy inside the wire contributes to the apparent inductive energy. In the presence of a vector potential on the interior of the wire, $\mathbf{A}^{\text{int}}$, a current density forms, and then Hamiltonian for the kinetic energy of the electrons is given by

$$H_{\text{el}}^{(T)} = \int_{\text{SC}} dV c_{rs}^\dagger \frac{\mathbf{p}_{\text{tot}}^2}{2m} c_{rs}. \quad (\text{F14})$$

We compute the kinetic energy with respect to the BCS ground state as

$$\begin{aligned}
 E_{\text{el}}^{(T)} &= \langle \Psi | H_{\text{el}}^{(T)} | \Psi \rangle \\
 &= \frac{m}{2D^2 e^2} \int_{\text{SC}} dV J^2 \langle \Psi | c_{rs}^\dagger c_{rs} | \Psi \rangle \\
 &= \frac{m}{De^2} \frac{H_0^2}{2\lambda_L^2 I_1^2(r_w/\lambda_L)} \cdot l \int_0^{2\pi} d\theta \int_0^{r_w} r dr I_0^2(r/\lambda_L) \\
 &= \frac{i_s^2}{2} \frac{m}{De^2} \frac{l}{4\pi\lambda_L^2} \frac{I_0^2(r_w/\lambda_L) - I_1^2(r_w/\lambda_L)}{I_1^2(r_w/\lambda_L)} \\
 &\approx \frac{1}{2} \frac{m}{De^2} \frac{l}{\sigma_{\text{eff}}} \frac{i_s^2}{2}, \quad \text{for } \lambda_L \ll r_w,
 \end{aligned} \tag{F15}$$

in which $\sigma_{\text{eff}} \approx 2\pi r_w \lambda_L$ and $H_0 = i_s/(2\pi r_w)$ from Eq. (F3).

Field energy inside the wire. In addition to the electron kinetic energy, an additional contribution to the kinetic inductance arises from the field energy on the interior of the superconductor, $E_{\text{B}^{\text{int}}}$:

$$\begin{aligned}
 E_{\text{B}^{\text{int}}} &= \frac{1}{2\mu_0} \int_{\text{SC}} dV |E_{\text{B}^{\text{int}}}|^2 \\
 &= \frac{\mu_0 i_s^2 l}{8\pi r_w I_1^2(r_w/\lambda_L)} \cdot l \int_0^{2\pi} d\theta \int_0^{r_w} r dr I_1^2(r/\lambda_L)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{i_s^2}{2} \frac{\mu_0 l}{4\pi r_w} \left[r_w - r_w \frac{I_0^2(r_w/\lambda_L)}{I_1^2(r_w/\lambda_L)} + 2\lambda_L \frac{I_0(r_w/\lambda_L)}{I_1(r_w/\lambda_L)} \right] \\
 &\approx \frac{i_s^2}{2} \frac{\mu_0 l \lambda_L}{4\pi r_w}, \quad \text{for } \lambda_L \ll r_w \\
 &= \frac{1}{2} \frac{m}{De^2} \frac{l}{\sigma_{\text{eff}}} \frac{i_s^2}{2},
 \end{aligned} \tag{F16}$$

where $\lambda_L = \sqrt{m/(\mu_0 De^2)}$. We see that $E_{\text{el}}^{(T)} = E_{\text{B}^{\text{int}}}$, and we find

$$E_{\text{int}} = E_{\text{el}}^{(T)} + E_{\text{B}^{\text{int}}} = L_K i_s^2/2, \tag{F17}$$

where the kinetic inductance L_K is given in Eq. (F10).

Total energy. Using Eqs. (F13) and (F17), the total energy of the system is given by

$$E_{\text{tot}} = E_{\text{ext}} + E_{\text{int}} = \underbrace{(L_G + L_K)}_{\equiv L} i_s^2/2, \tag{F18}$$

from which we see that the total (energetic) inductance of the system is in agreement with the inductance computed from the integrated flux, given in Eq. (F12).

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